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(54) **POST CMP PROCESSING FOR HYBRID BONDING**

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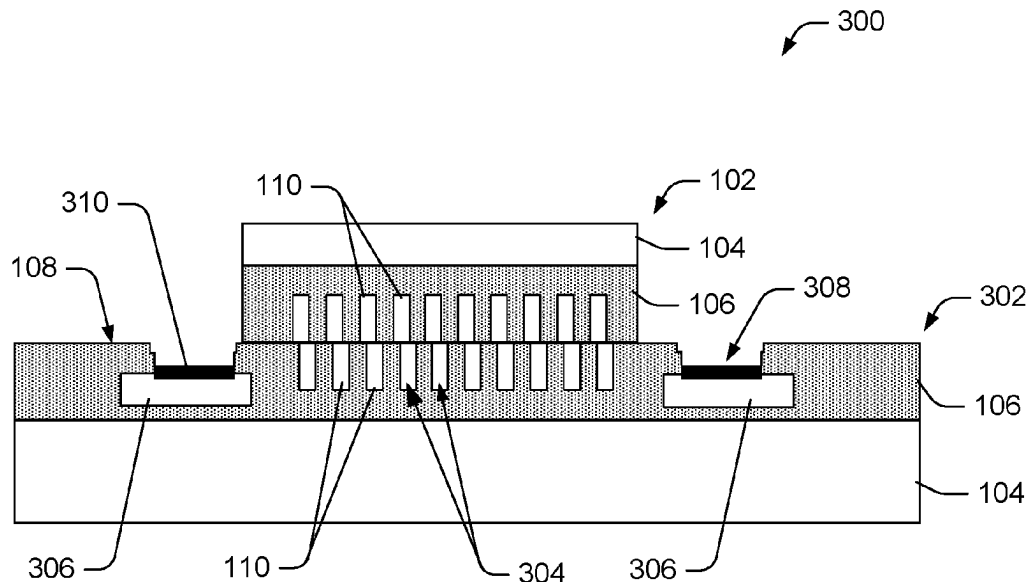
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(57) **ABSTRACT**

Devices and techniques include process steps for forming openings through stacked and bonded structures. The openings are formed by pre-etching through one or more layers of prepared dies after planarization of the bonding layer (by chemical-mechanical polishing (CMP) or the like) and prior to bonding. For instance, the openings are etched through one or more layers of dies to be bonded prior to bonding the dies to form an assembly.

20 Claims, 28 Drawing Sheets



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 Onsemi AR0820 image, cross section of a CMOS image sensor product. The part in the image was shipped on Sep. 16, 2021. Applicant makes no representation that the part in the image is

identical to the part identified in the separately submitted reference Bush, Nov. 8, 2018, *Electronics Weekly.com* ("Bush article"); however, the imaged part and the part shown in the Bush article share the part number "Onsemi AR0820."

Sony IMX260 image, cross section of Sony dual-pixel sensor product labeled IMX260, showing peripheral probe and wire bond pads in a bonded structure. The part in the image was shipped in Apr. 2016. Applicant makes no representation that the part in the image is identical to the part identified in the separately submitted reference Morrison et al. (Tech Insights article dated Apr. 24, 2016), describing and showing a similar sensor product within the Samsung Galaxy S7; however the imaged part and the part shown in the Morrison et al. article share the part name "Sony IMX260."

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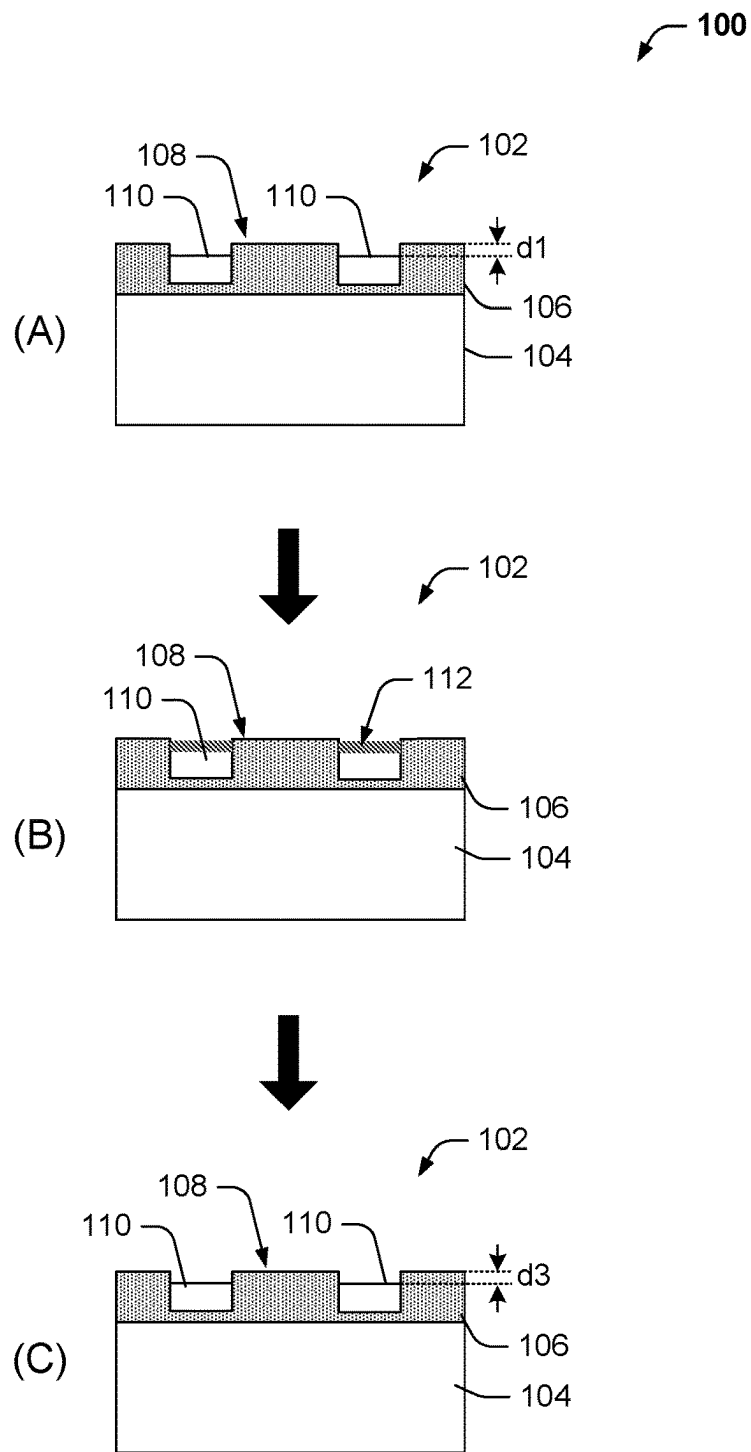
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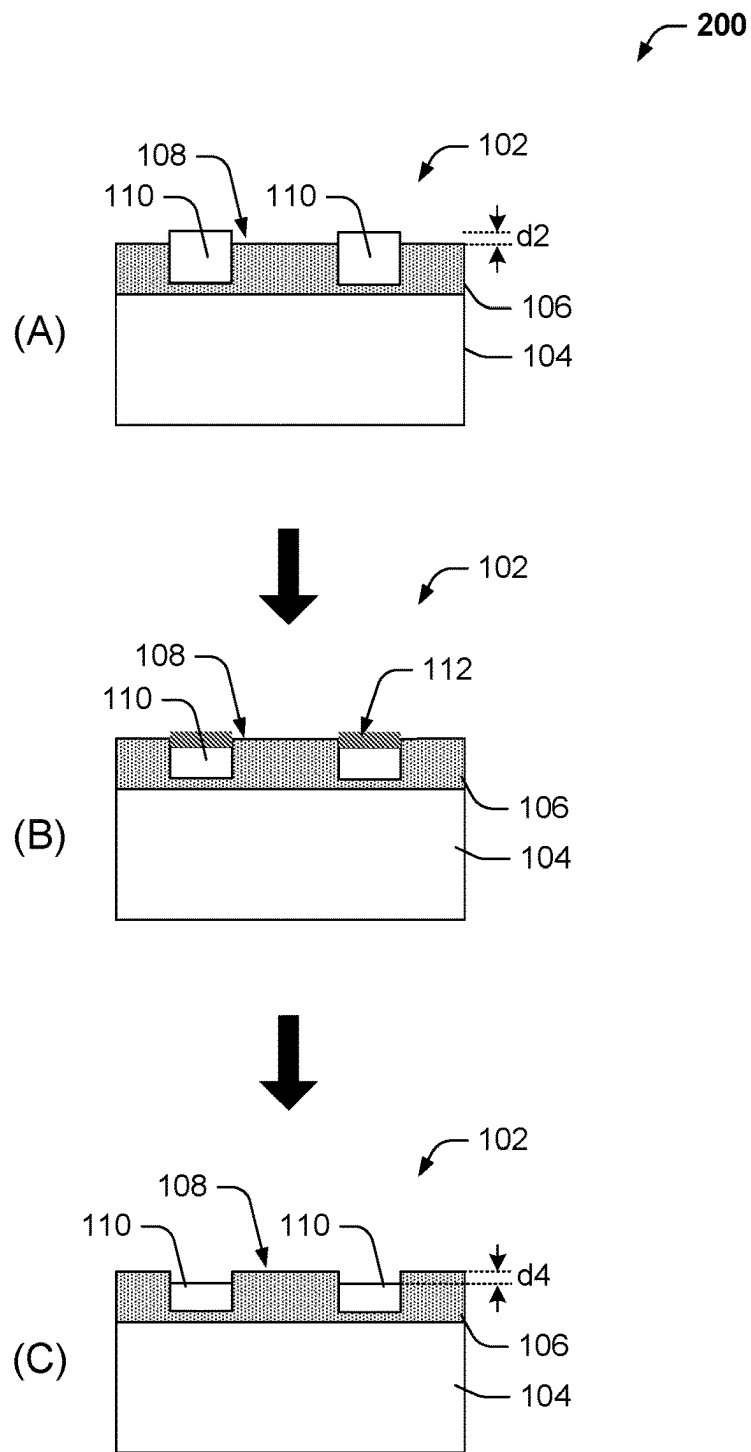


FIG. 2

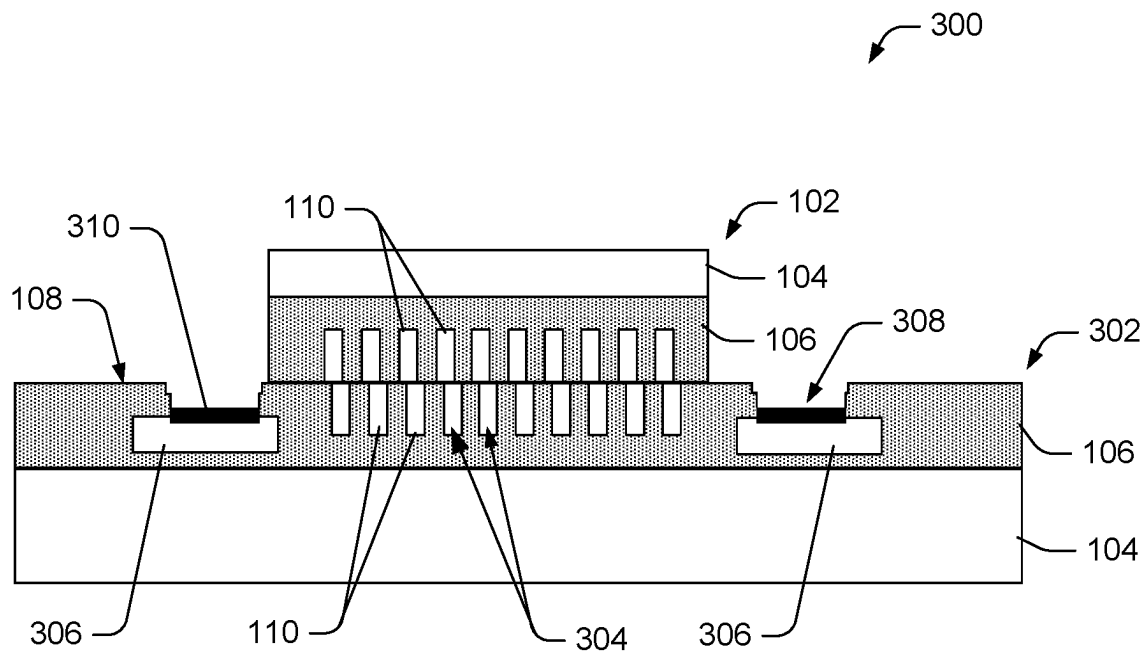


FIG. 3

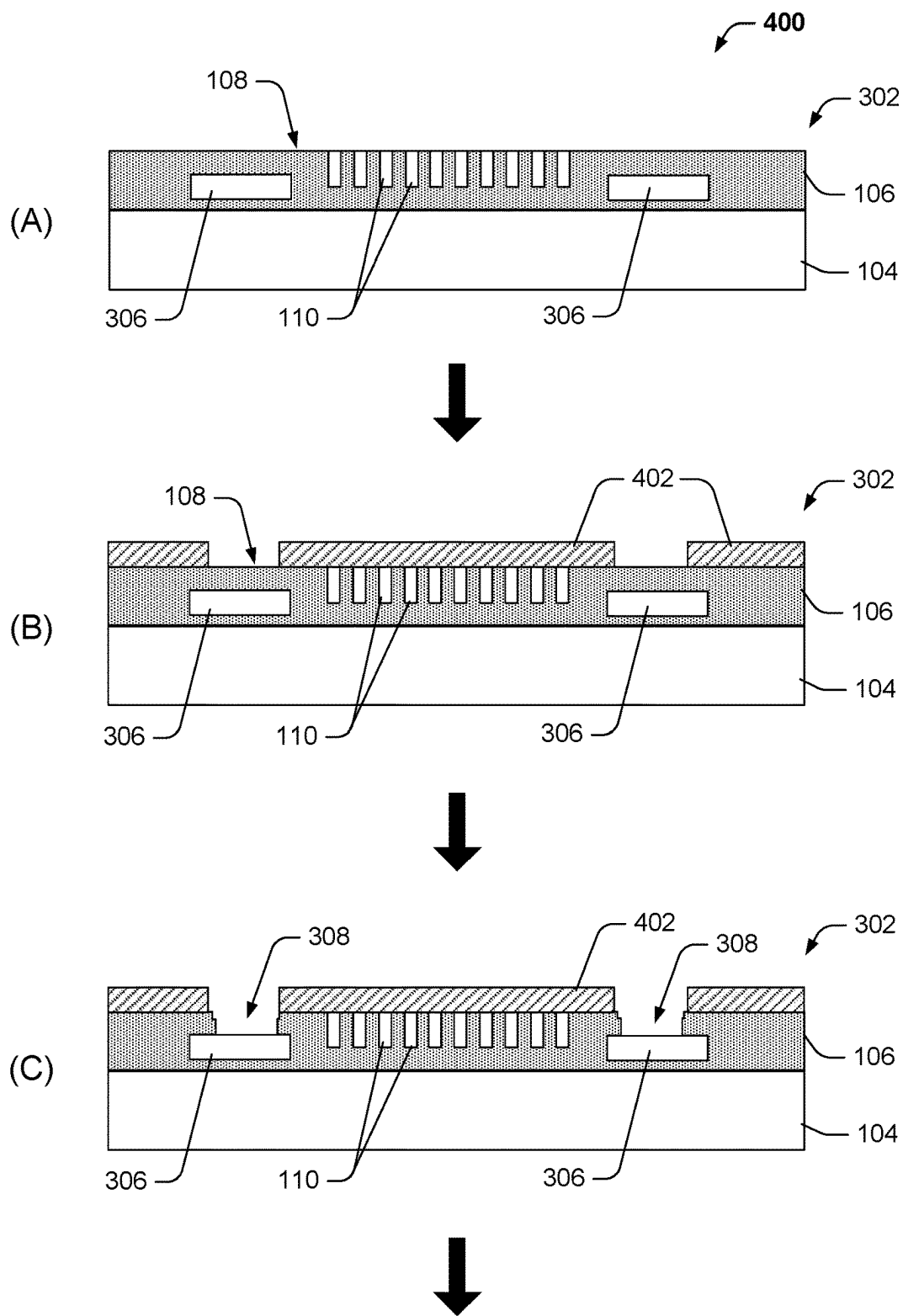
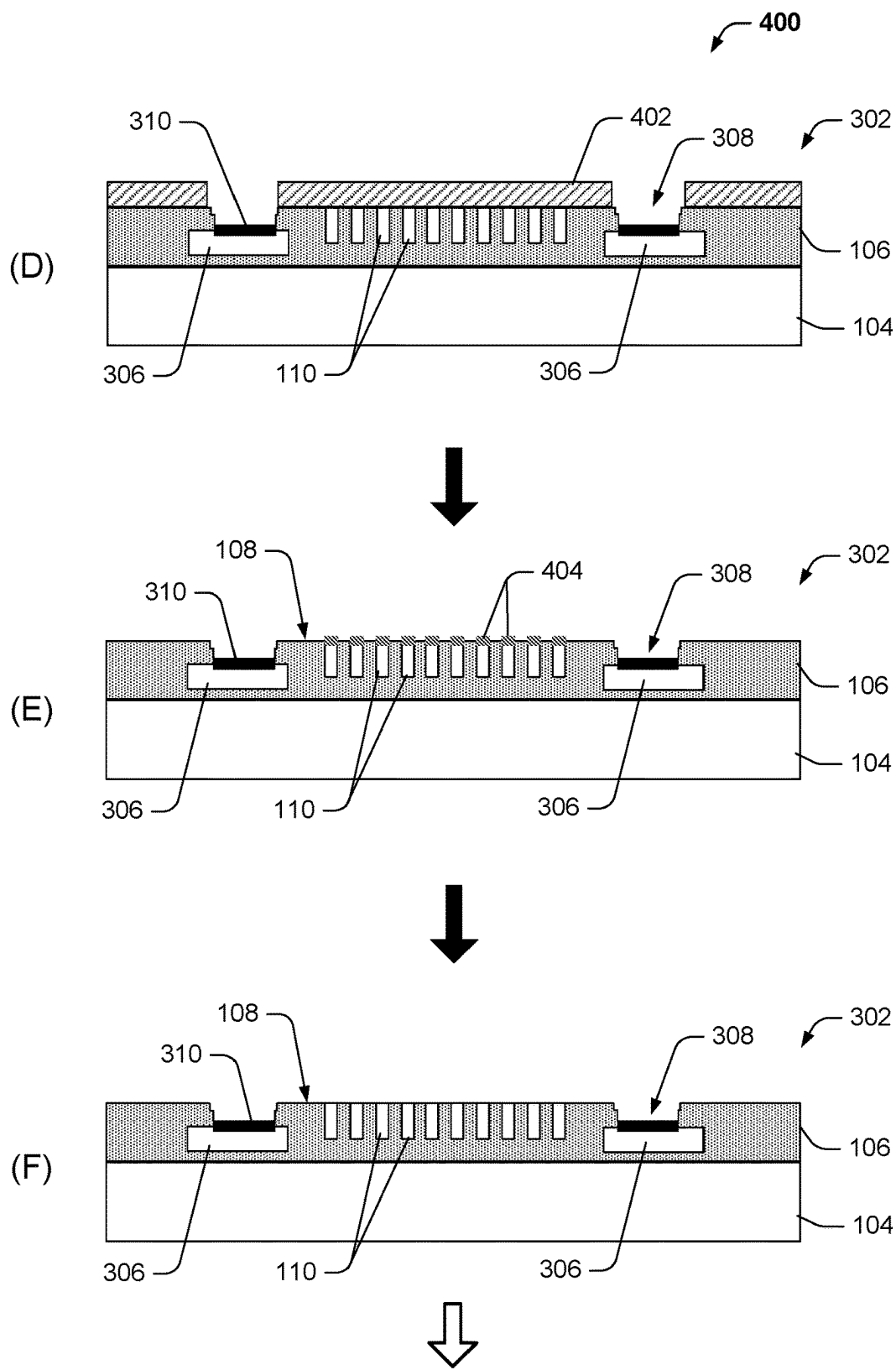


FIG. 4



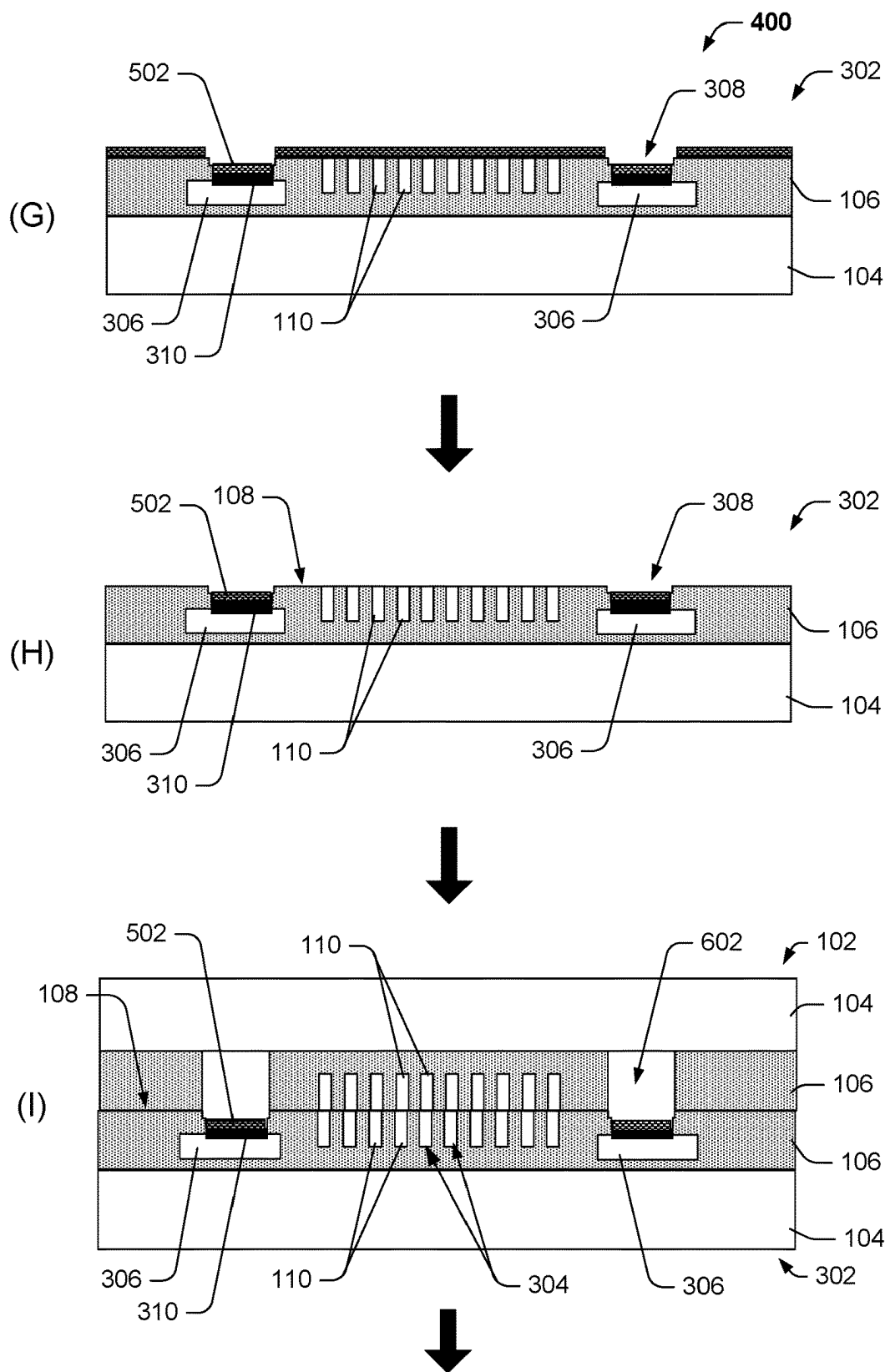
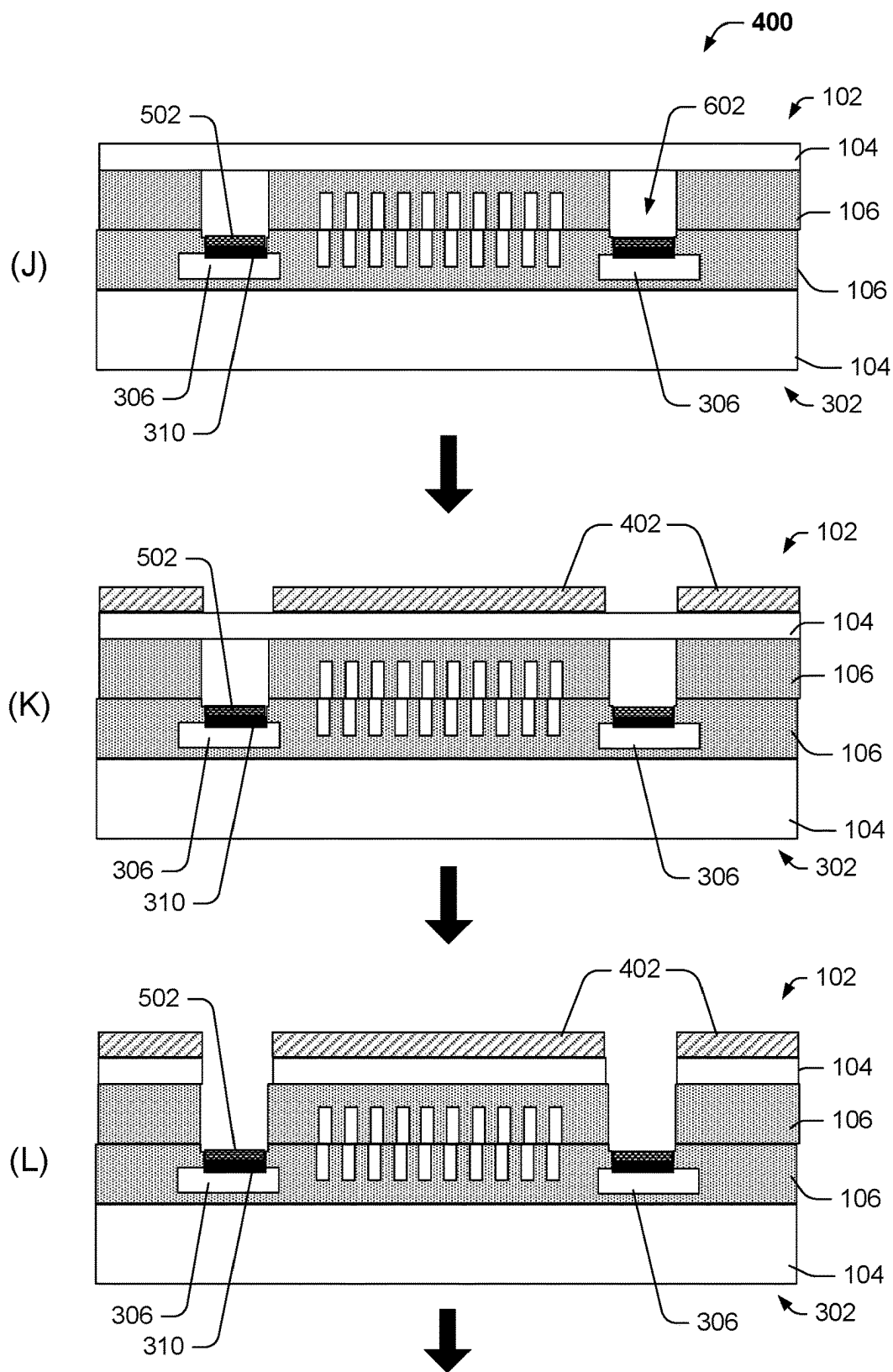


FIG. 5A



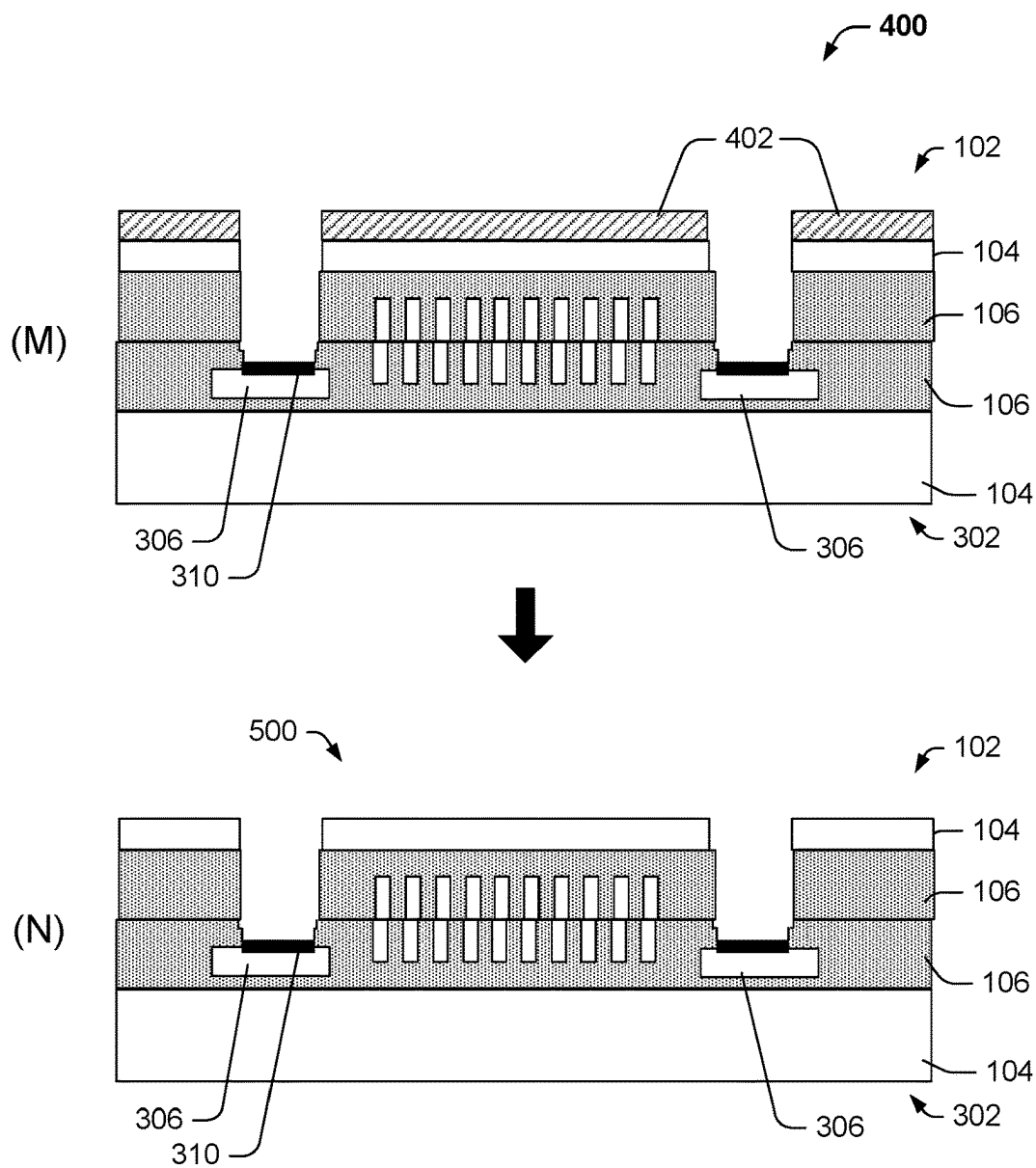


FIG. 5C

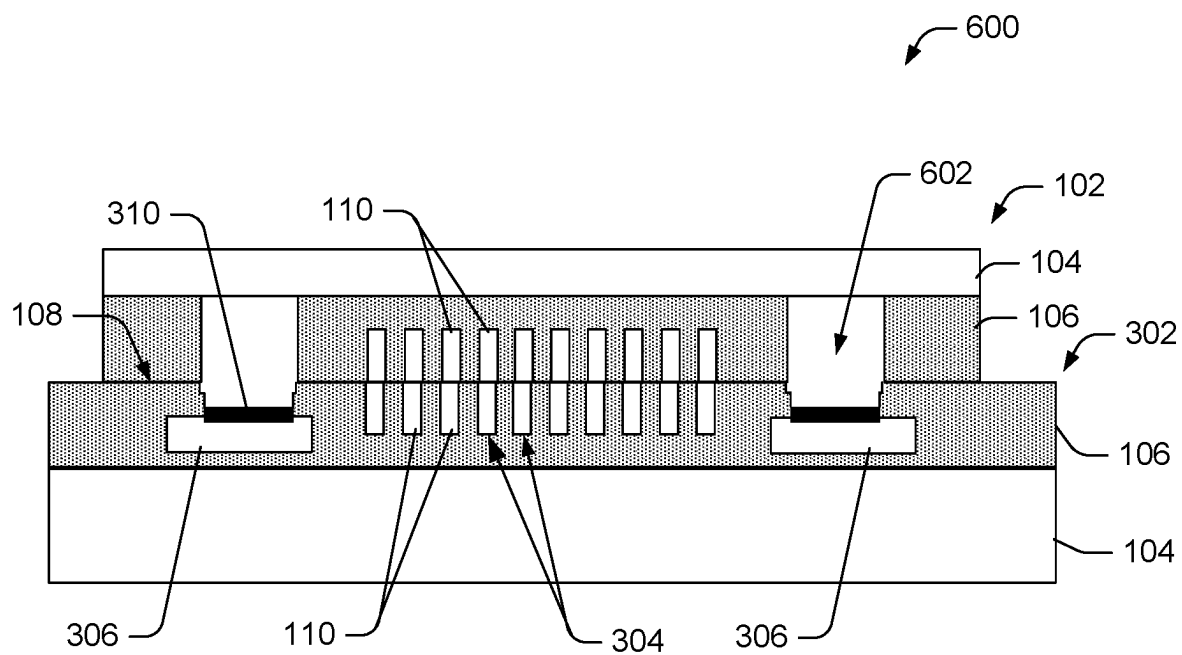


FIG. 6

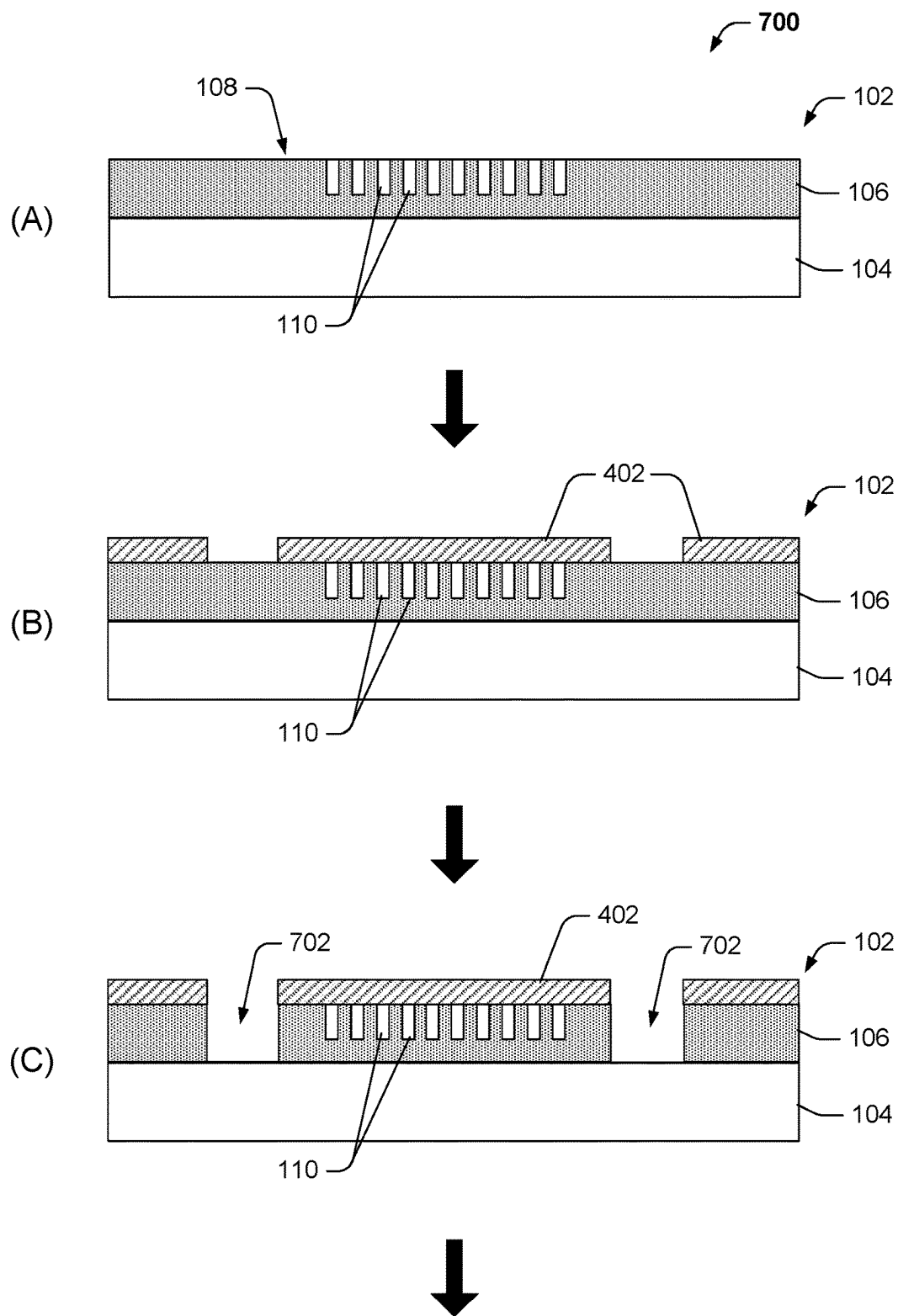


FIG. 7

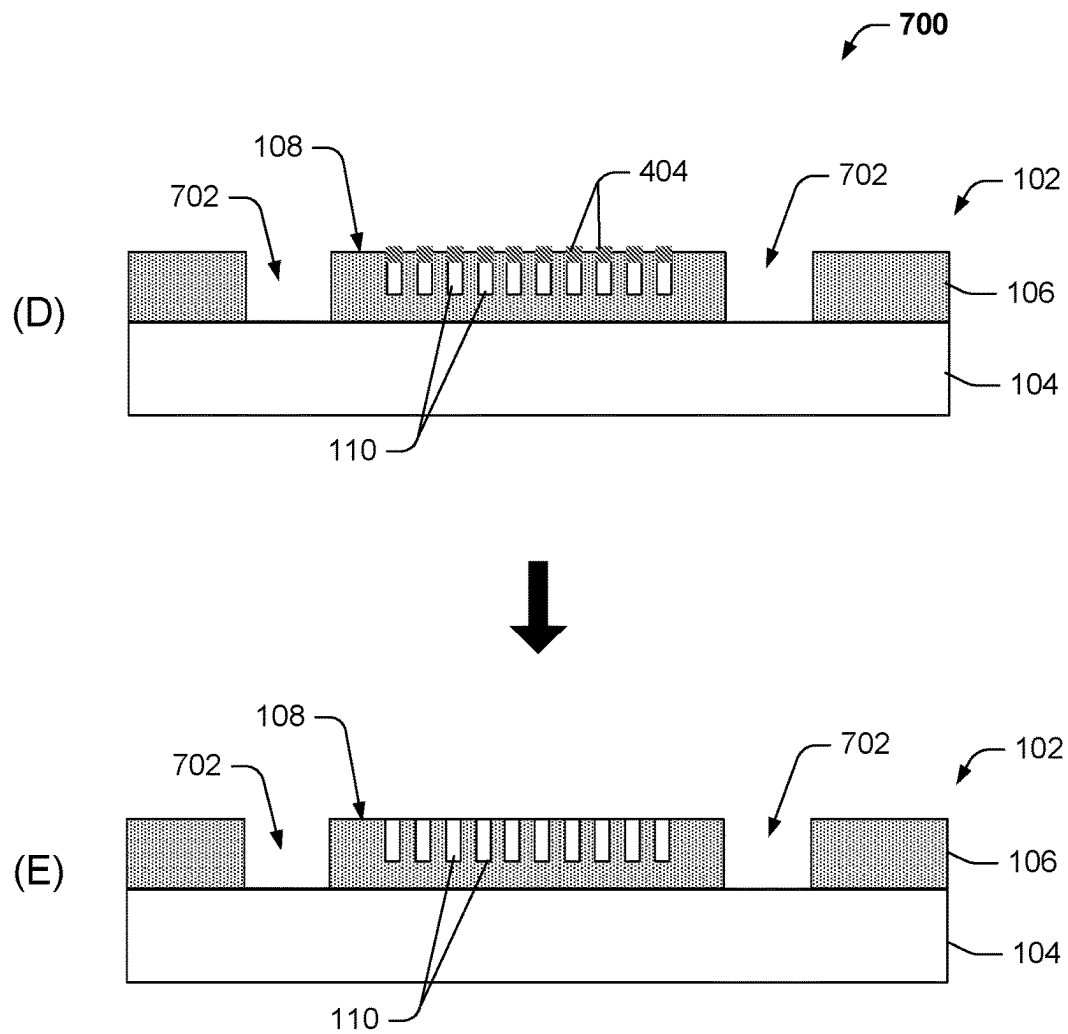


FIG. 8

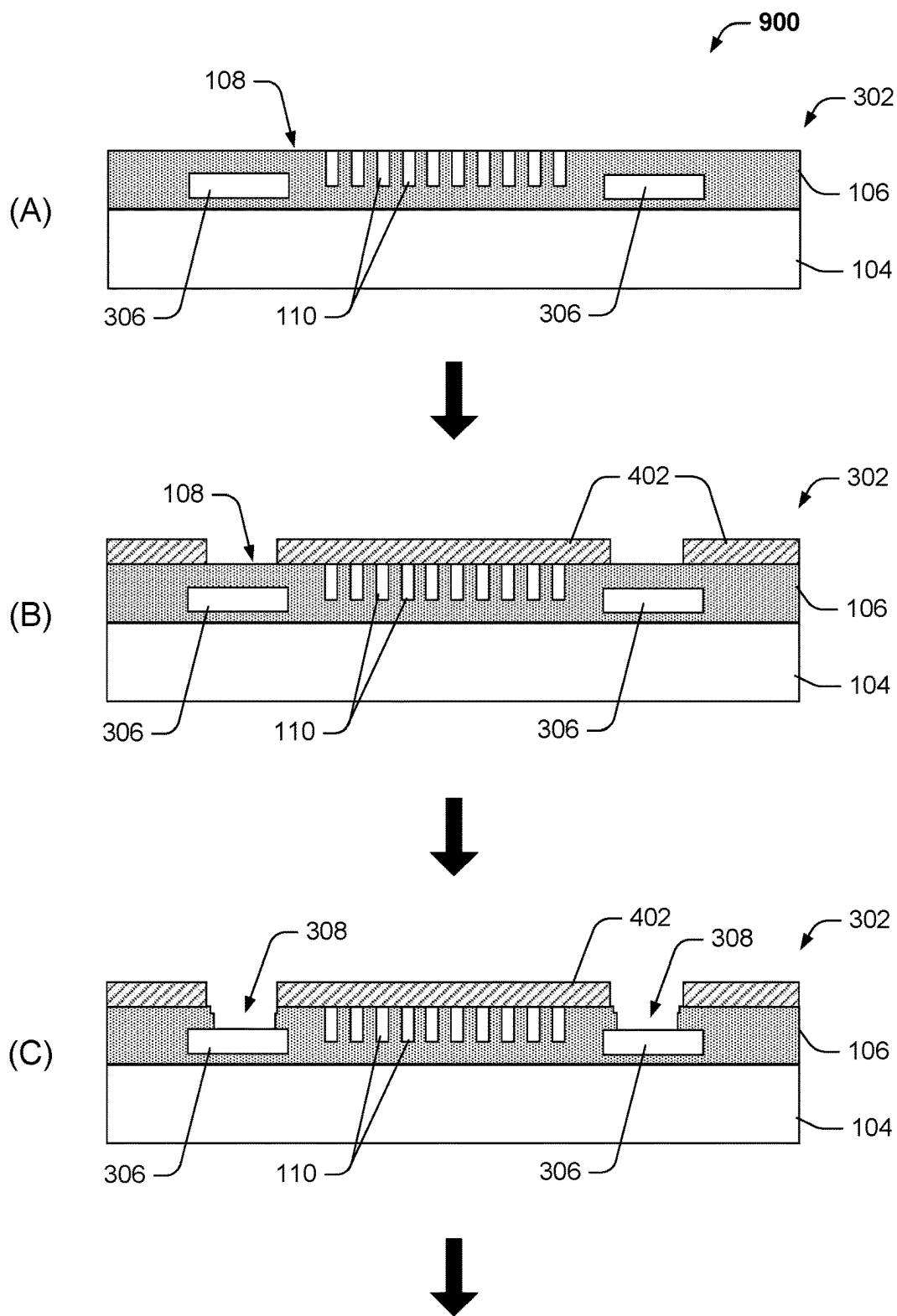
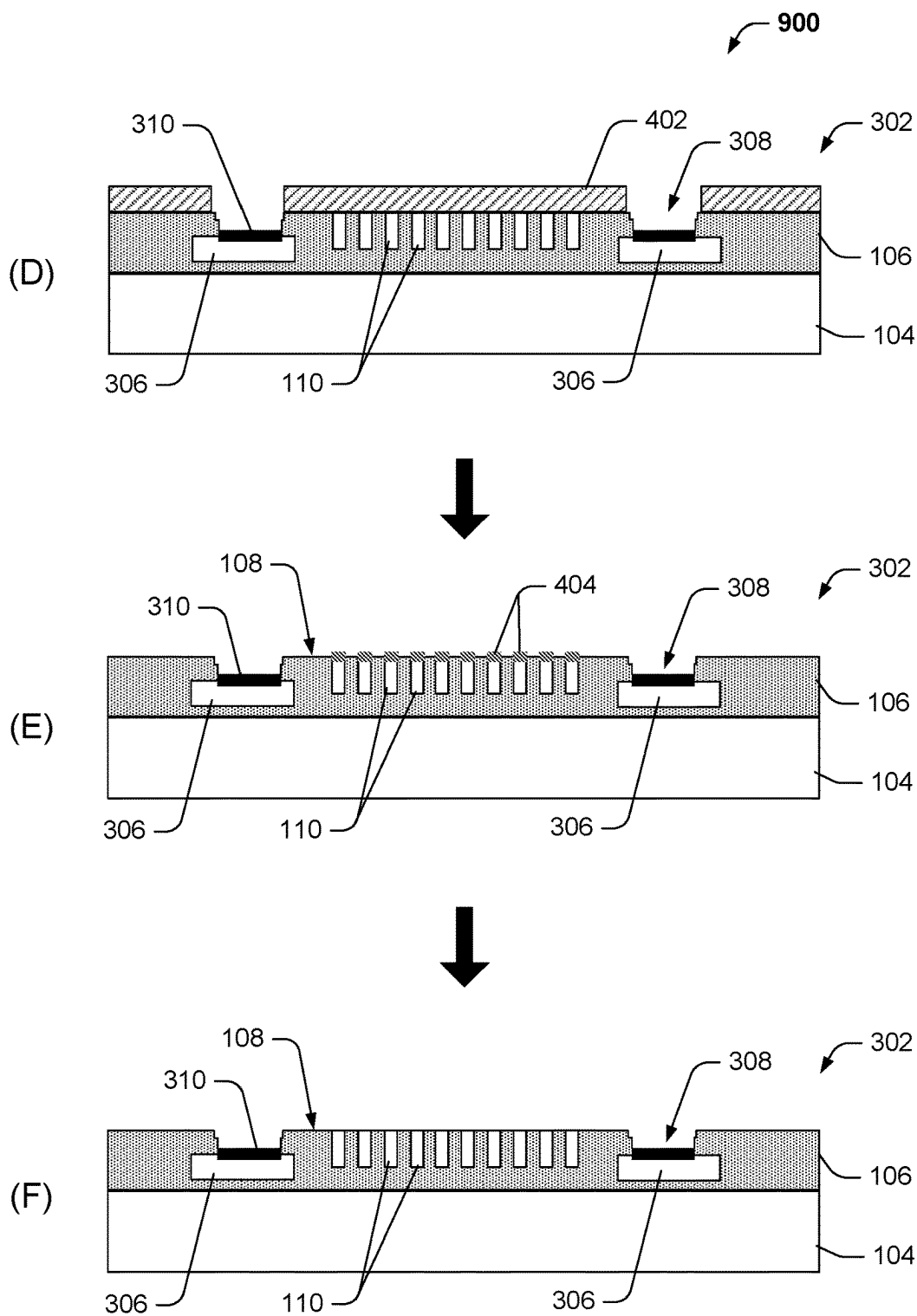
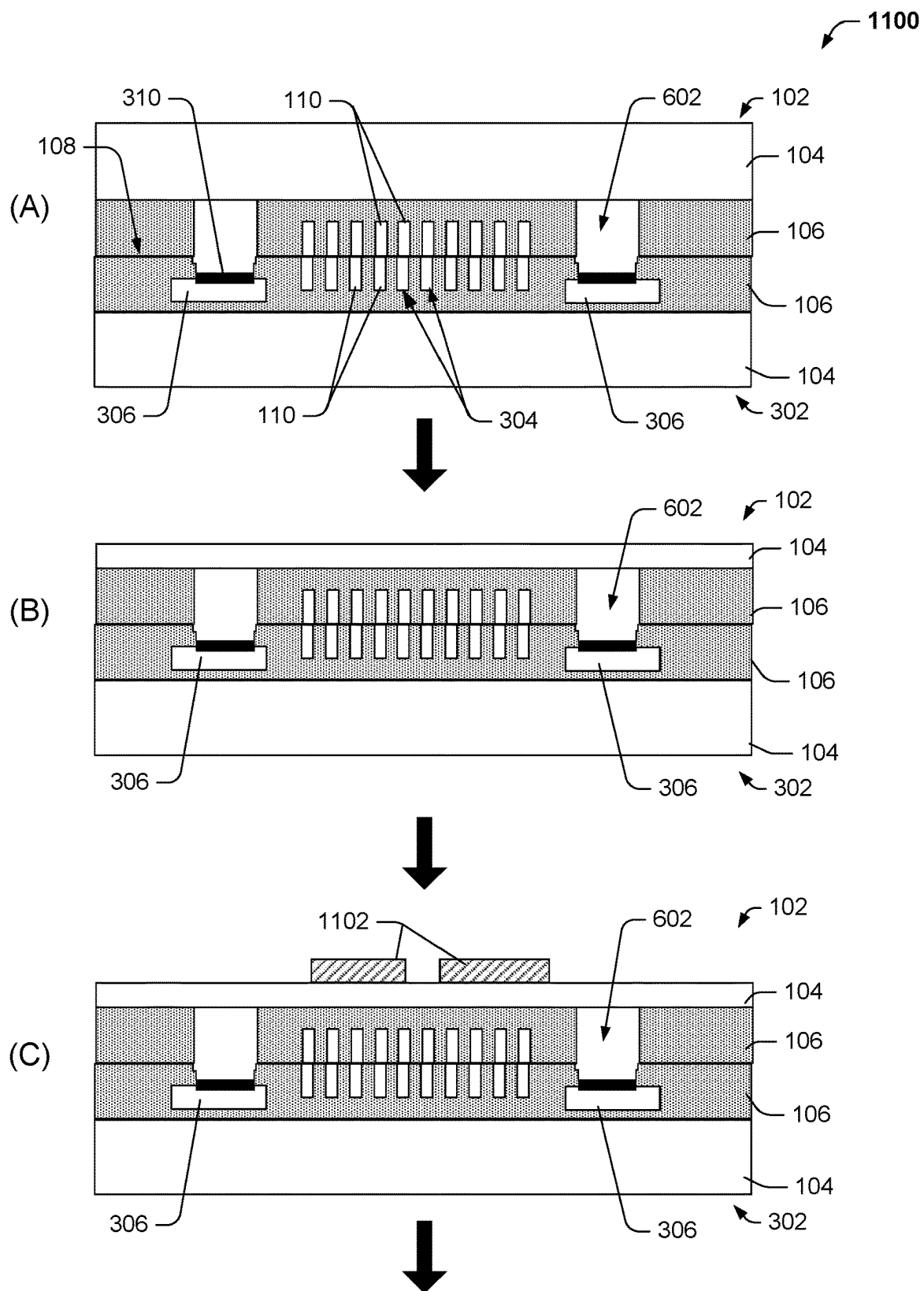


FIG. 9





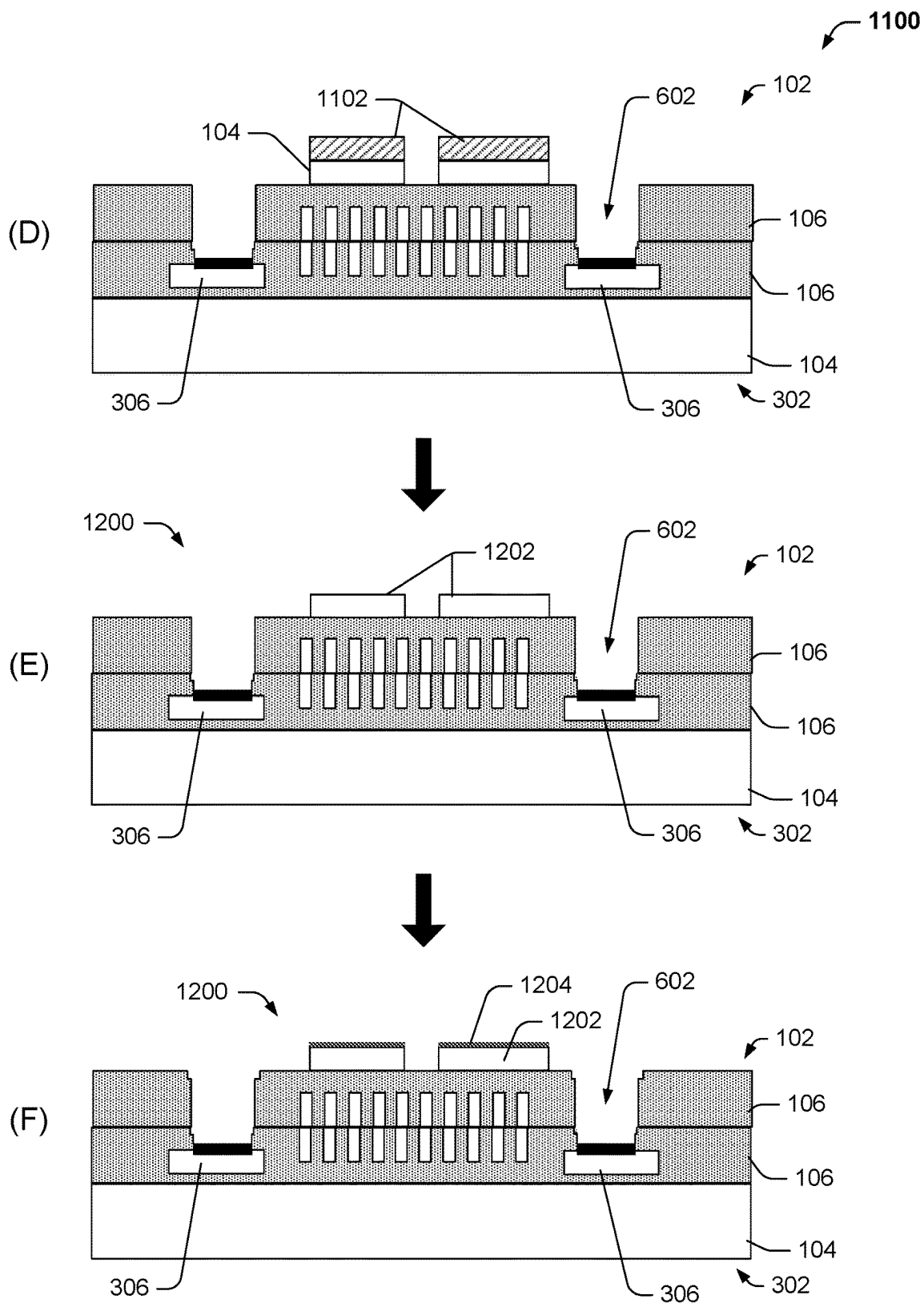


FIG. 12

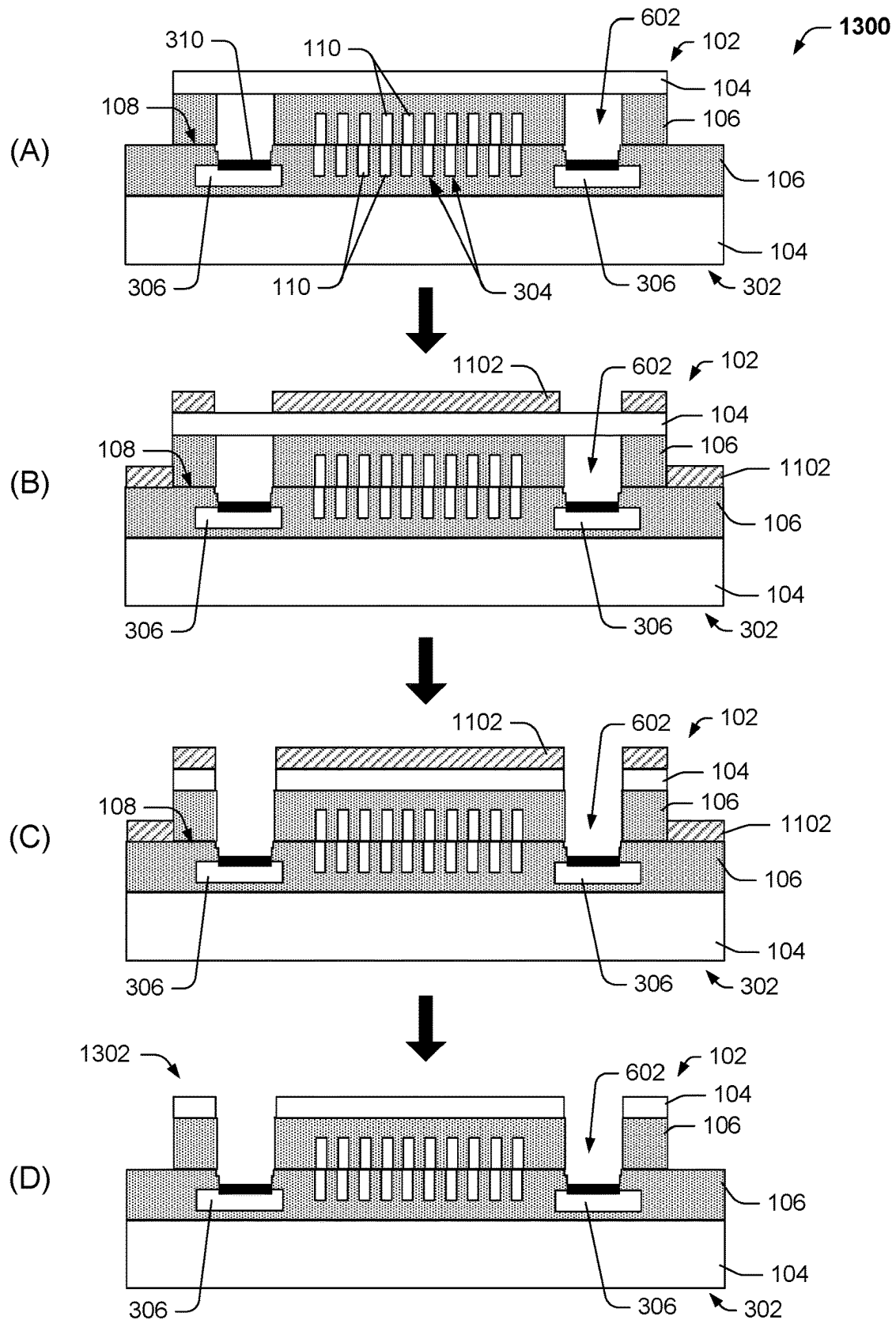


FIG. 13

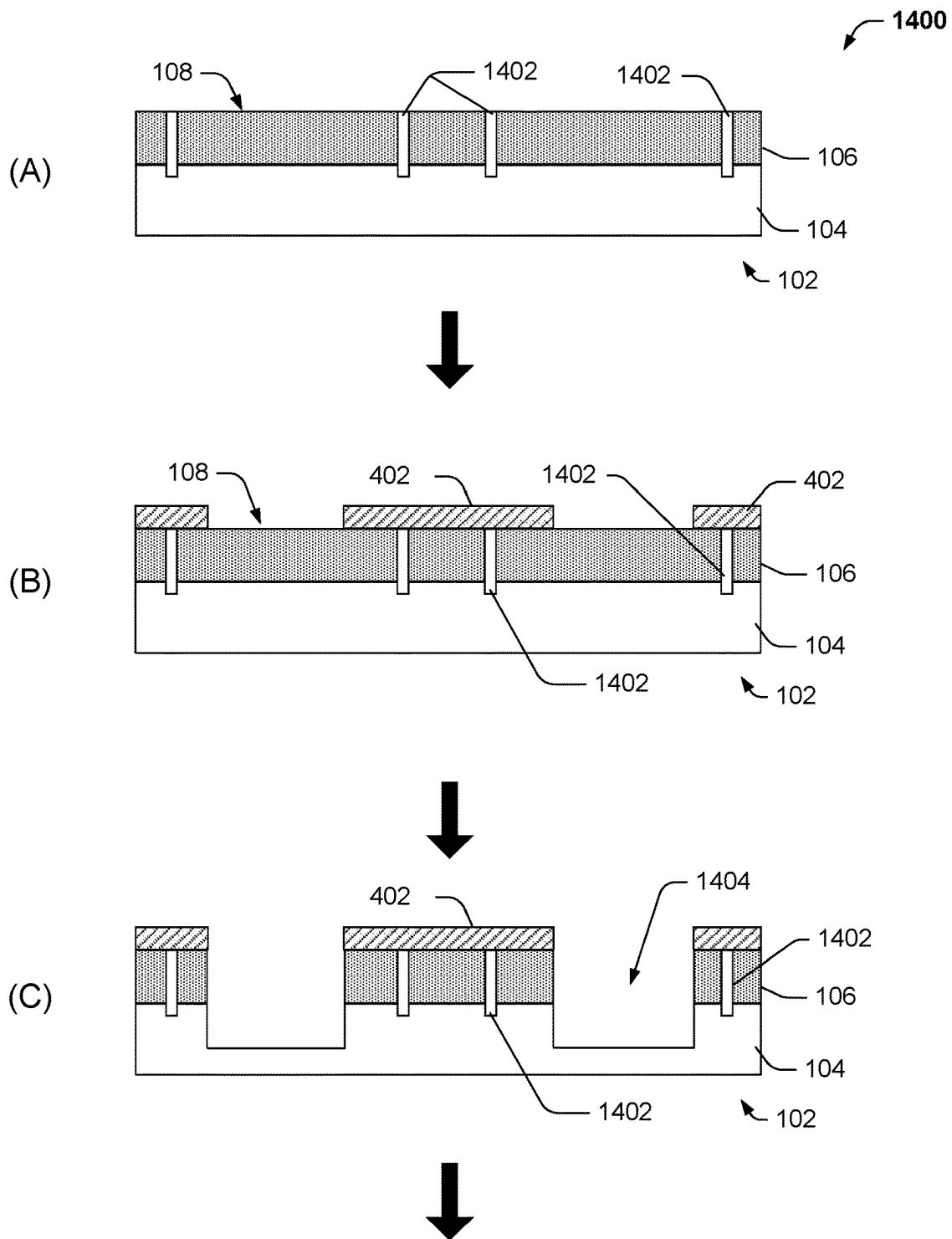


FIG. 14

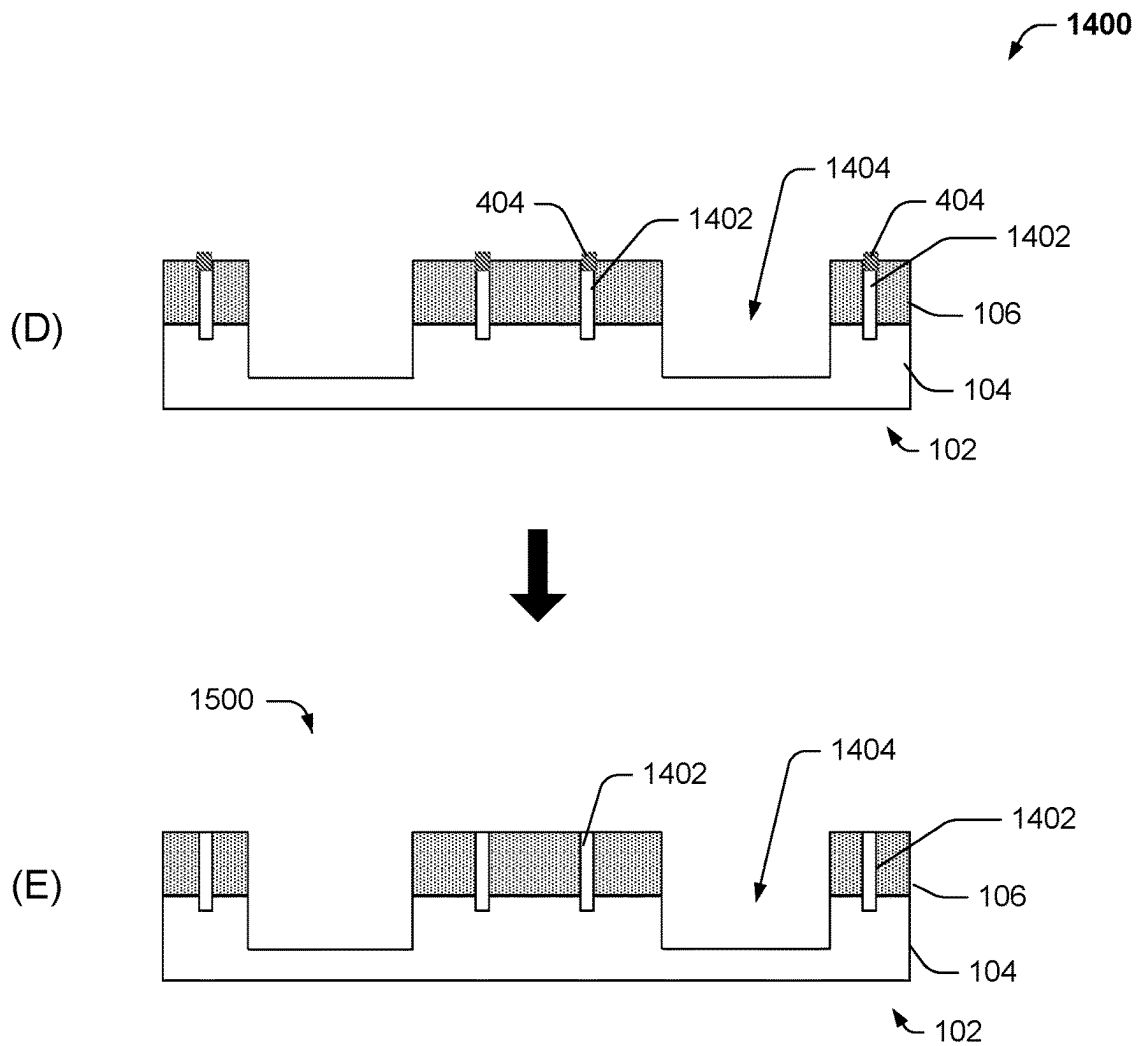


FIG. 15

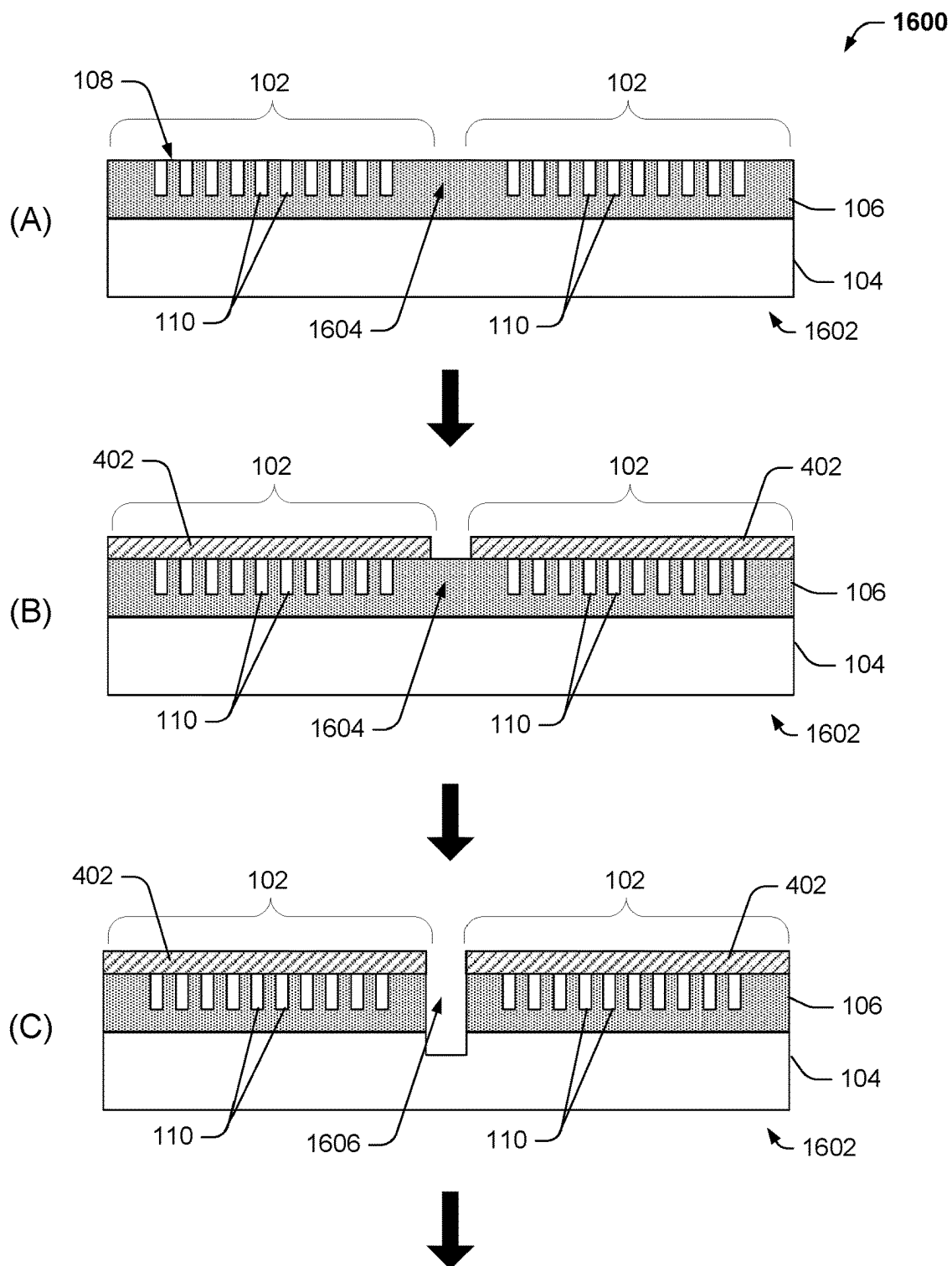


FIG. 16

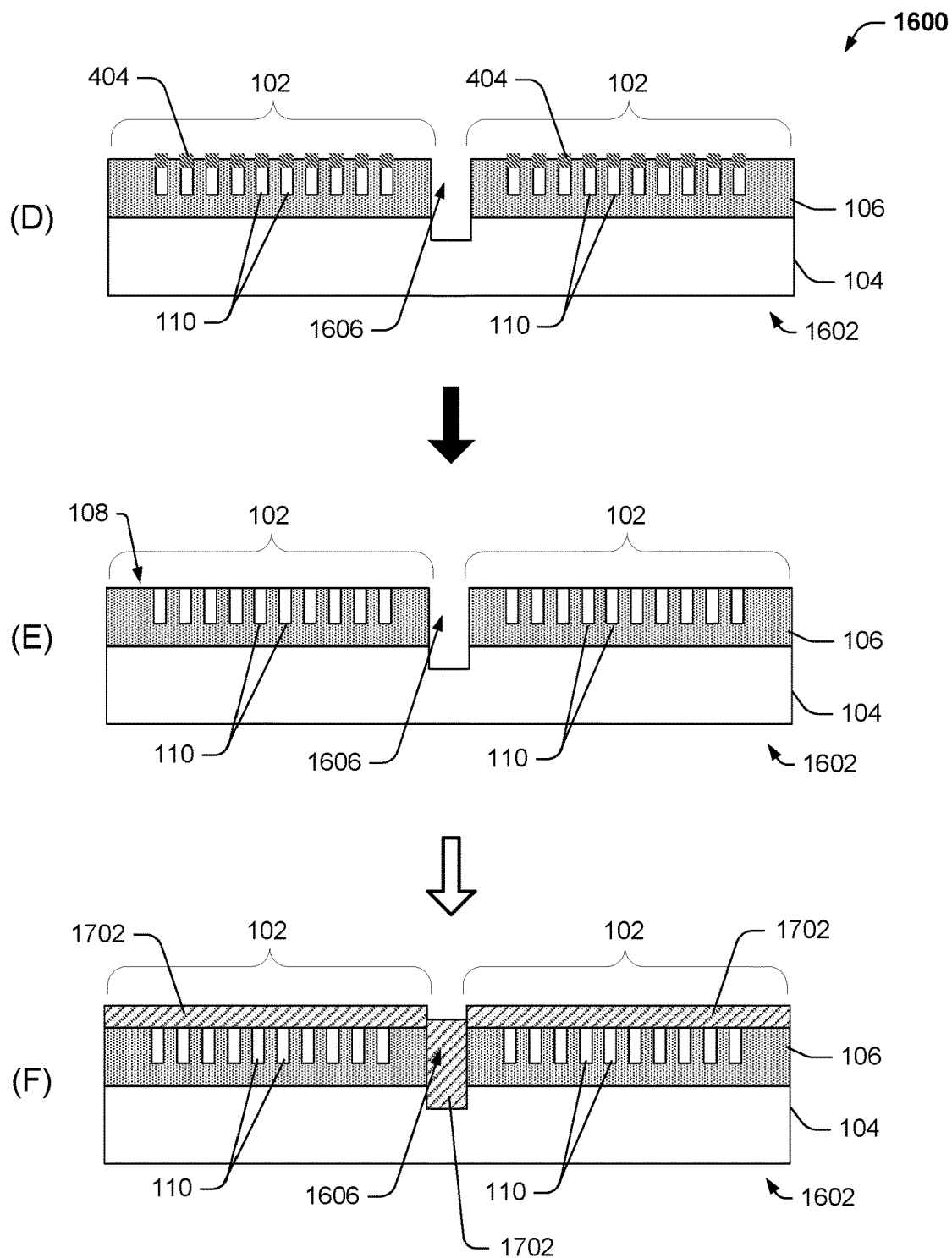


FIG. 17

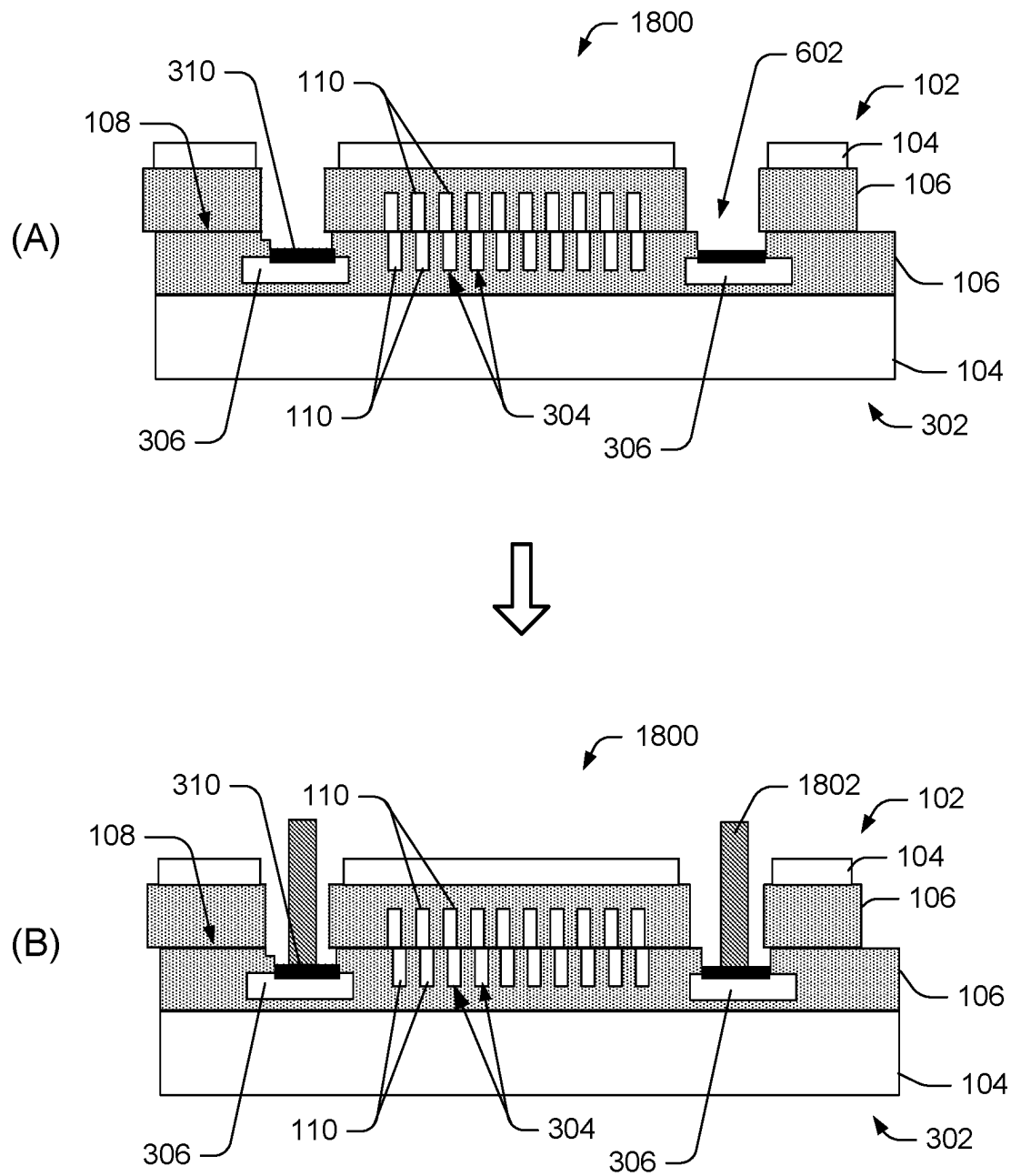


FIG. 18

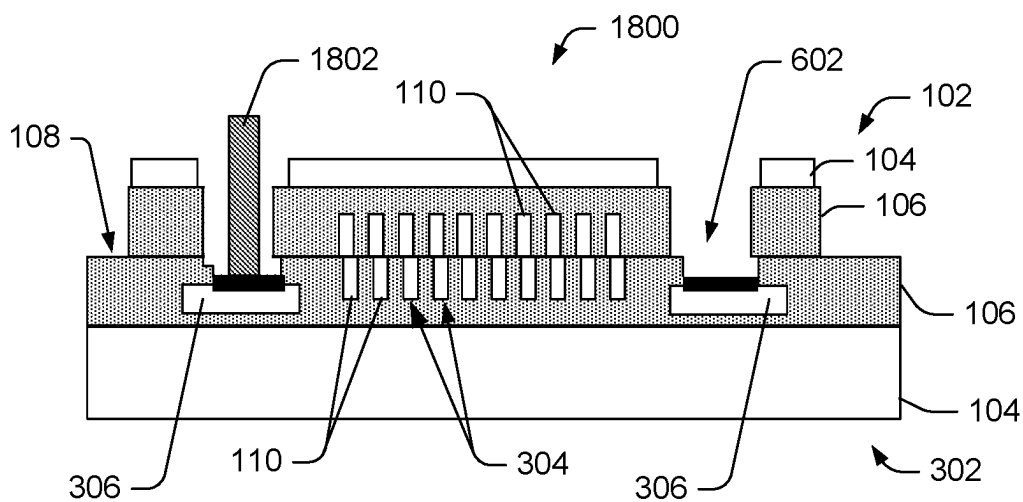


FIG. 19

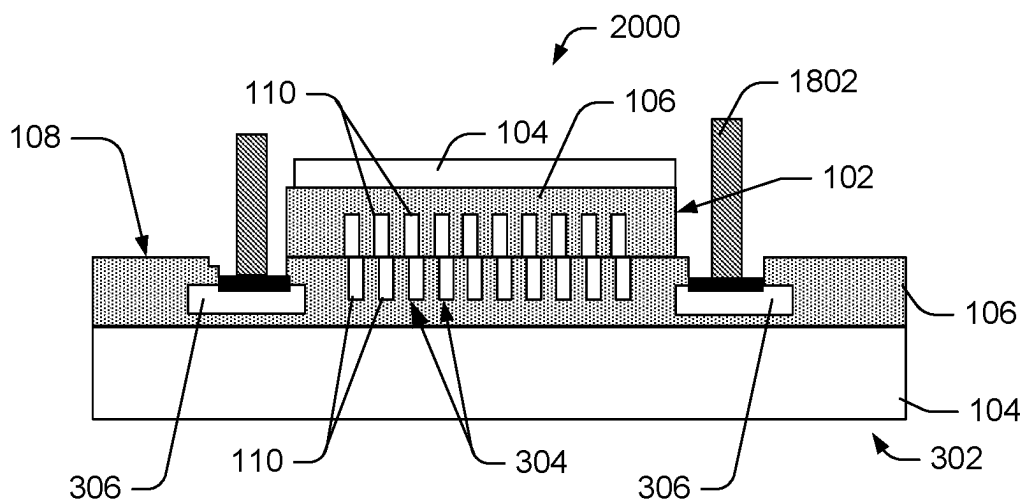


FIG. 20

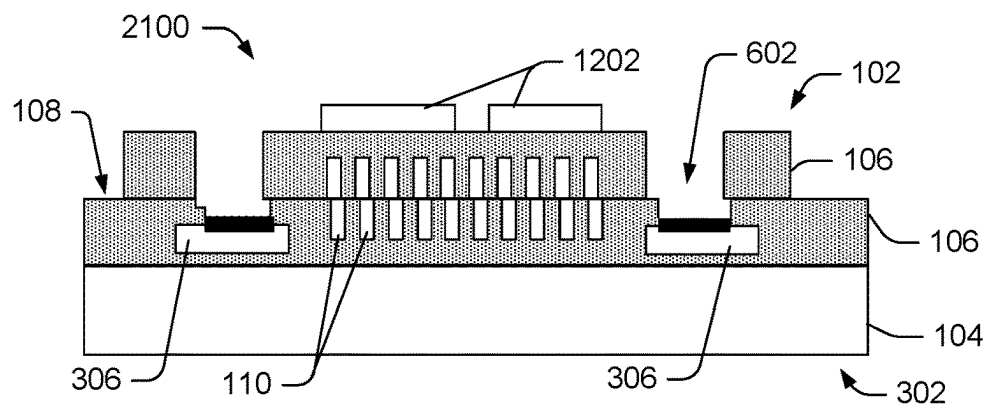


FIG. 21

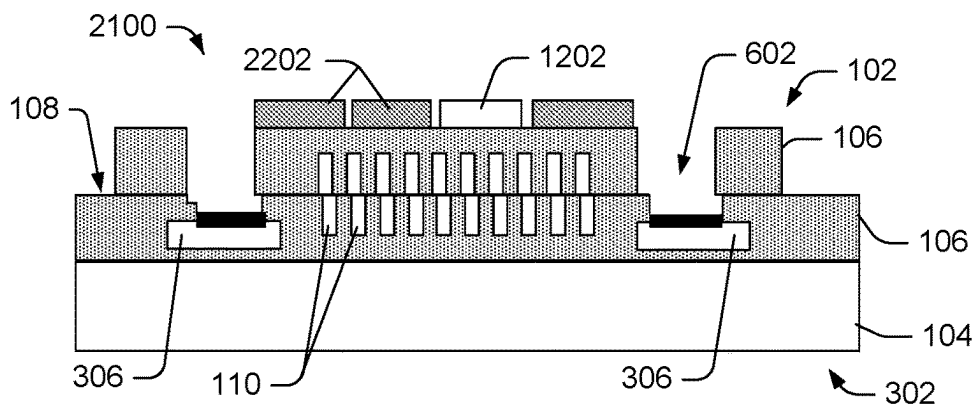


FIG. 22

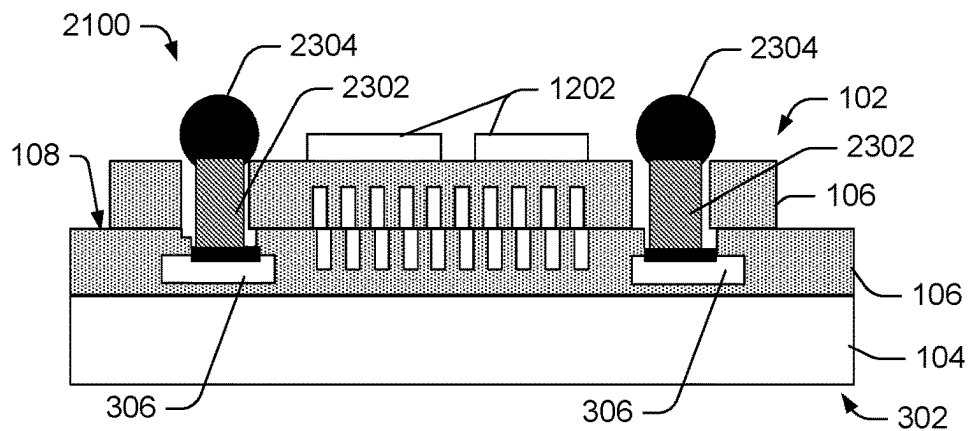


FIG. 23

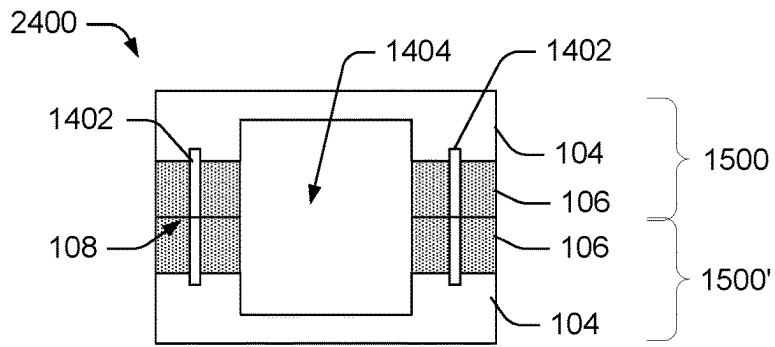


FIG. 24

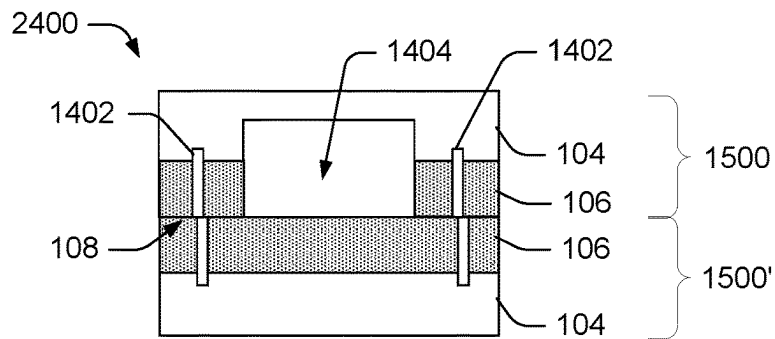


FIG. 25

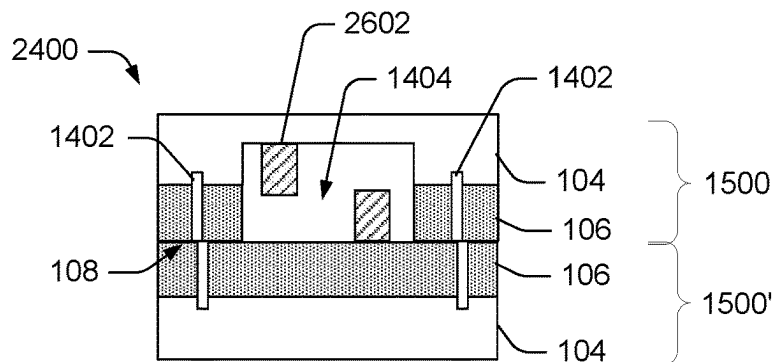


FIG. 26

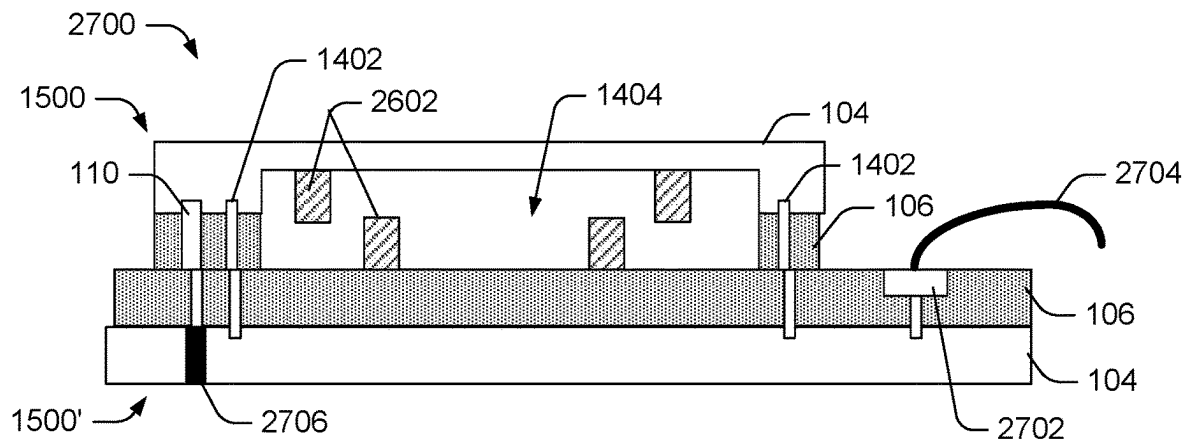


FIG. 27

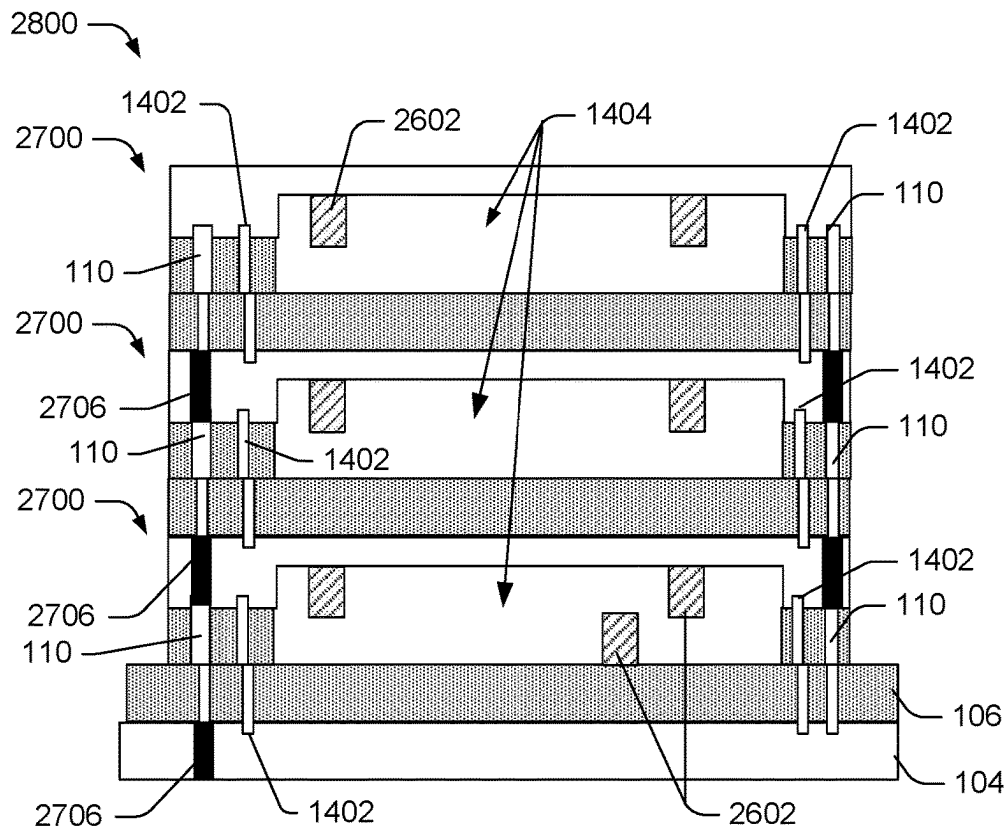


FIG. 28

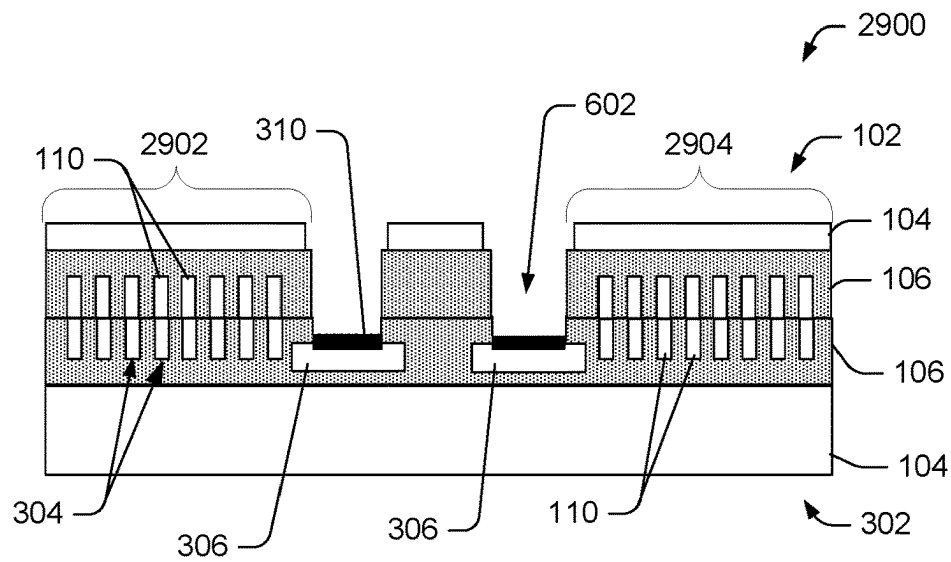


FIG. 29

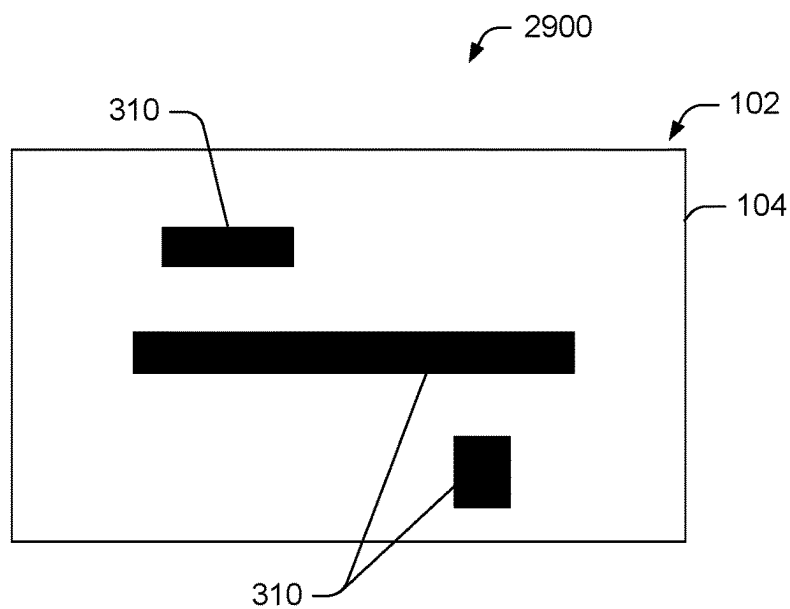


FIG. 30

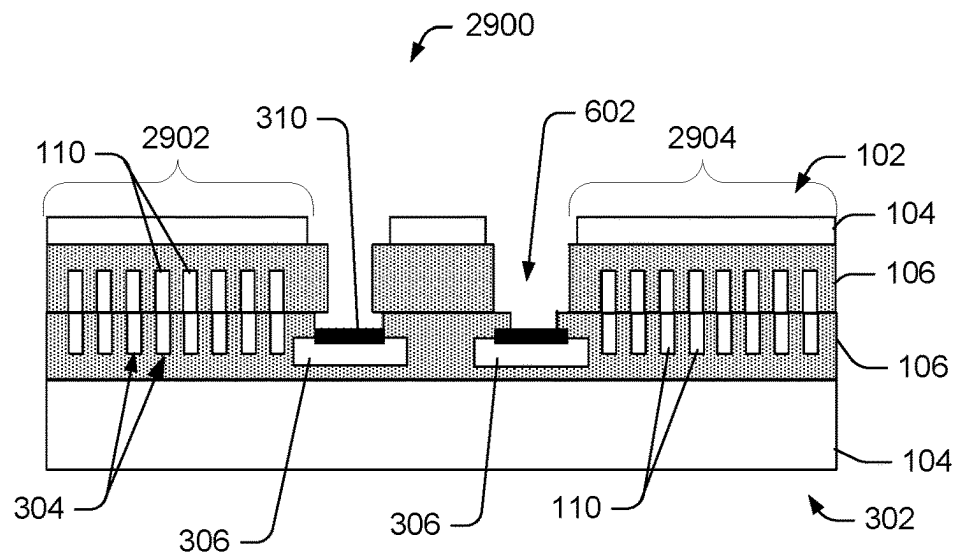


FIG. 31

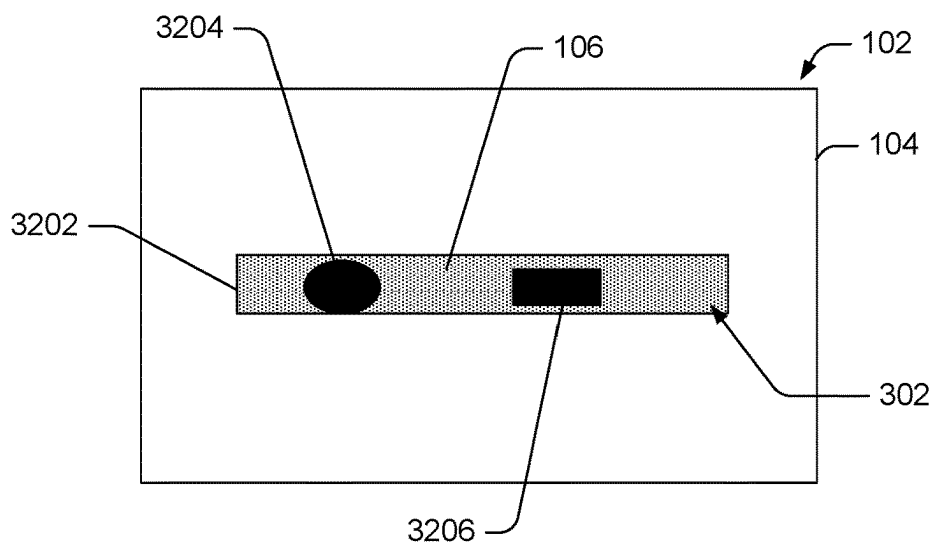


FIG. 32

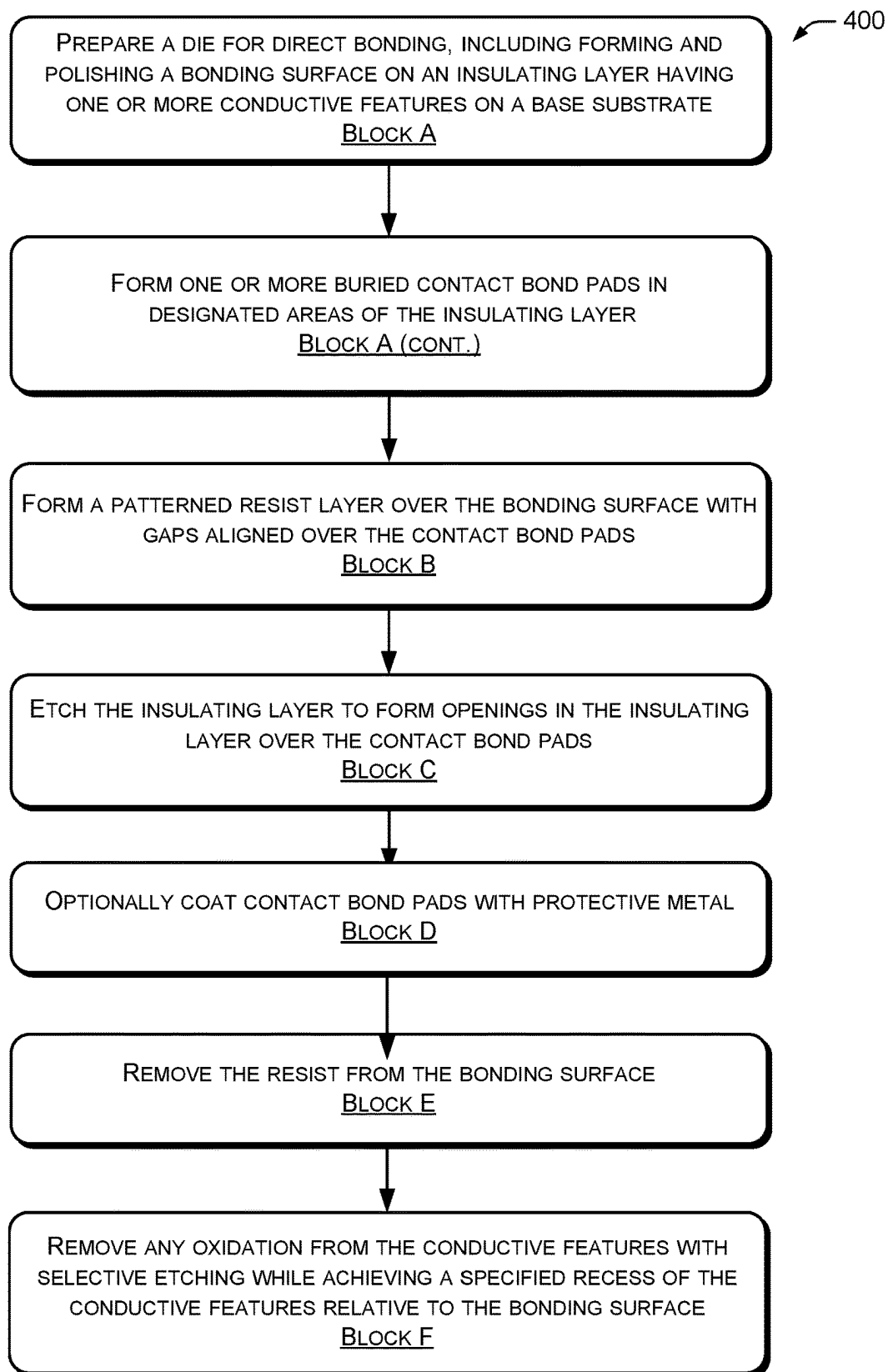


FIG. 33

1

POST CMP PROCESSING FOR HYBRID BONDING

PRIORITY CLAIM AND CROSS-REFERENCE TO RELATED APPLICATION

This application claims the benefit under 35 U.S.C. § 119(e)(1) of U.S. Provisional Application No. 62/703,727, filed Jul. 26, 2018, which is hereby incorporated by reference in its entirety.

FIELD

The following description relates to integrated circuits ("ICs"). More particularly, the following description relates to manufacturing IC dies and wafers.

BACKGROUND

Microelectronic elements often comprise a thin slab of a semiconductor material, such as silicon or gallium arsenide, commonly called a semiconductor wafer. A wafer can be formed to include multiple integrated chips or dies on a surface of the wafer and/or partly embedded within the wafer. Dies that are separated from a wafer are commonly provided as individual, prepackaged units. In some package designs, the die is mounted to a substrate or a chip carrier, which is in turn mounted on a circuit panel, such as a printed circuit board (PCB). For example, many dies are provided in packages suitable for surface mounting.

Packaged semiconductor dies can also be provided in "stacked" arrangements, wherein one package is provided, for example, on a circuit board or other carrier, and another package is mounted on top of the first package. These arrangements can allow a number of different dies or devices to be mounted within a single footprint on a circuit board and can further facilitate high-speed operation by providing a short interconnection between the packages. Often, this interconnect distance can be only slightly larger than the thickness of the die itself. For interconnection to be achieved within a stack of die packages, interconnection structures for mechanical and electrical connection may be provided on both sides (e.g., faces) of each die package (except for the topmost package).

Additionally, dies or wafers may be stacked in a three-dimensional arrangement as part of various microelectronic packaging schemes. This can include stacking a layer of one or more dies, devices, and/or wafers on a larger base die, device, wafer, substrate, or the like, stacking multiple dies or wafers in a vertical or horizontal arrangement, and various combinations of both.

Dies or wafers may be bonded in a stacked arrangement using various bonding techniques, including direct dielectric bonding, non-adhesive techniques, such as ZiBond® or a hybrid bonding technique, such as DBI®, both available from Invensas Bonding Technologies, Inc. (formerly Ziptronix, Inc.), an Xperi company (see for example, U.S. Pat. Nos. 6,864,585 and 7,485,968, which are incorporated herein in their entirety). Respective mating surfaces of the bonded dies or wafers often include embedded conductive interconnect structures, or the like. In some examples, the bonding surfaces are arranged and aligned so that the conductive interconnect structures from the respective surfaces are joined during the bonding. The joined interconnect structures form continuous conductive interconnects (for signals, power, etc.) between the stacked dies or wafers.

2

There can be a variety of challenges to implementing stacked die and wafer arrangements. When bonding stacked dies using a direct bonding or hybrid bonding technique, it is usually desirable that the surfaces of the dies to be bonded be extremely flat, smooth, and clean. For instance, in general, the surfaces should have a very low variance in surface topology (i.e., nanometer scale variance), so that the surfaces can be closely mated to form a lasting bond.

Conductive interconnect structures at the bonding surfaces may be slightly recessed, just below the insulating material of the bonding surface. The amount of recess below the bonding surface may be determined by a dimensional tolerance, specification, or physical limitation of the device or application. The hybrid surface may be prepared for bonding with another die, wafer, or other substrate using a chemical mechanical polishing (CMP) process, or the like.

Additionally, in some applications it is desired to form wirebond pads, testing pads, and other structures on or beside a bonding surface, or within cavities at the bonding surface, to be accessed, often from outside of the stacked and bonded device, after bonding. Openings or cavities may be used to create access ports for sensor applications, physical and electrical access (e.g., for wirebonding, testing, electrical connections, etc.), low-impedance, low-loss connections, air bridges, and so forth. Forming cavities to accommodate these structures, connections, accesses, and for other purposes, through stacked and bonded dies, wafers, and substrates can be problematic, particularly when the cavities desired are to have a fine pitch and/or a significant depth.

BRIEF DESCRIPTION OF THE DRAWINGS

The detailed description is set forth with reference to the accompanying figures. In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical items.

For this discussion, the devices and systems illustrated in the figures are shown as having a multiplicity of components. Various implementations of devices and/or systems, as described herein, may include fewer components and remain within the scope of the disclosure. Alternatively, other implementations of devices and/or systems may include additional components, or various combinations of the described components, and remain within the scope of the disclosure.

FIGS. 1 and 2 show example graphical flow diagrams illustrating example processes for preparing a bonding surface, according to an embodiment.

FIG. 3 shows an example assembly formed from stacked and bonded devices, having hybrid bonding pads and peripheral pads, according to an embodiment.

FIGS. 4 and 5 illustrate a graphical flow diagram showing an example process for forming one of the devices of the assembly of FIG. 3, according to an embodiment.

FIGS. 5A-5C illustrate optional steps to the example process described at FIGS. 4 and 5, according to an embodiment.

FIG. 6 shows an example assembly formed from stacked and bonded devices, having hybrid bonding pads and peripheral pads inside the die perimeter, according to an embodiment.

FIGS. 7 and 8 illustrate a graphical flow diagram showing an example process for forming one of the devices of the assembly of FIG. 6, according to an embodiment.

FIGS. 9 and 10 illustrate a graphical flow diagram showing an example process for forming another of the devices of the assembly of FIG. 6, according to an embodiment.

FIGS. 11 and 12 illustrate a graphical flow diagram showing an example process for forming an example assembly formed from stacked and bonded devices, having hybrid bonding pads and peripheral pads inside the die perimeter, according to an embodiment.

FIG. 13 illustrates a graphical flow diagram showing an example process for forming an example assembly formed from stacked and bonded devices, having hybrid bonding pads and peripheral pads inside the die perimeter, according to an embodiment.

FIGS. 14 and 15 illustrate a graphical flow diagram showing an example process for forming a cavity with a seal ring or contact hybrid bonding surface, according to an embodiment.

FIGS. 16 and 17 illustrate a graphical flow diagram showing an example process for street etching a wafer prior to singulation, according to an embodiment.

FIGS. 18-23 show examples of assemblies formed from stacked and bonded devices, having hybrid bonding pads and peripheral pads, according to various embodiments.

FIGS. 24-26 show examples of assemblies formed from stacked and bonded devices, having interior cavities and seal rings, according to various embodiments.

FIG. 27 shows an example of an assembly formed from stacked and bonded devices, having hybrid bonding pads and seal rings, and at least one peripheral pad outside the die perimeter, according to an embodiment.

FIG. 28 shows an example of an assembly formed from multiple stacked and bonded devices, having hybrid bonding pads and seal rings, according to an embodiment.

FIGS. 29-32 show example stacked and bonded microelectronic assemblies having etched openings through layers of the microelectronic assemblies, according to various embodiments.

FIG. 33 is a flow diagram illustrating an example process for preparing dies for bonding, according to an embodiment.

SUMMARY

Representative techniques and devices are disclosed, including process steps for forming holes, cavities, openings, recesses, and the like (hereinafter "openings") through stacked and bonded structures. Of particular interest are intimately bonded structures, formed using direct bonding and hybrid bonding techniques without adhesive, as discussed above. In various embodiments, the openings are formed by pre-etching through one or more layers of prepared wafers or dies after planarization of the bonding layer (by chemical-mechanical polishing (CMP) or the like) and prior to bonding. For instance, the openings are etched through one or more layers of dies to be bonded prior to bonding the dies to form an assembly.

In various implementations, a microelectronic assembly comprises a first substrate having a bonding surface with a planarized topography of inorganic dielectric material or insulator, and a first plurality of electrically conductive features embedded in the dielectric layer with one surface exposed at the bonding surface of the first substrate, and a second substrate having a bonding surface with a planarized topography of inorganic dielectric material or insulator, bonded to the bonding surface of the first substrate. The second substrate includes a second plurality of electrically

substrate, bonded to the first plurality of electrically conductive features. The first and second pluralities of electrically conductive features are direct-bonded, for instance, using a hybrid bonding technique or the like, without the use of solder or other added bonding materials. The use of the term "conductive features" will be used herein to specifically refer to these direct-bond conductive features.

One or more electrically conductive contact pads, separate from the direct-bond electrically conductive features, are disposed within an insulating layer of the second substrate and below the bonding surface of the second substrate. The one or more electrically conductive contact pads may be disposed outside or inside a perimeter of the first plurality of electrically conductive features and the second plurality of electrically conductive features or in one or more designated areas. The electrically conductive contact pads may include any conductive structure other than the electrically conductive features, which may be used as conductive interconnects for wirebonds, terminals, test pads, ball grids, and so forth.

Where the "conductive features" of one die or wafer may have matching or mating conductive features on a second direct-bonded die or wafer, the "contact pads" are unmatched for the purposes of die-to-die, wafer-to-wafer, or die-to-wafer direct bonding. For example, the contact pads of one die or wafer generally do not have a mating contact pad to be directly bonded to on the second bonded die or wafer. Further, the contact pads are generally on a different (e.g., "lower") layer of a die or wafer than the conductive features, and may be disposed or buried below the bonding surface of the die or wafer.

In an implementation, the microelectronic assembly further comprises one or more secondary openings in the insulating layer of the second substrate aligned to the one or more electrically conductive contact pads. The one or more secondary openings extend from the bonding surface of the second substrate to the one or more electrically conductive contact pads, providing access to the one or more electrically conductive contact pads.

In another implementation, the microelectronic assembly further comprises one or more primary openings in an insulating layer of the first substrate, aligned to the one or more secondary openings and to the electrically conductive contact pads. The one or more primary openings extend to the one or more secondary openings, providing access to the one or more electrically conductive contact pads.

In another implementation, the microelectronic assembly further comprises one or more tertiary openings in a base layer of the first substrate, aligned to the one or more primary openings in the insulating layer of the first substrate and to the electrically conductive contact pads. The one or more tertiary openings extend from an outside surface of the first substrate to the one or more primary openings, providing access to the one or more electrically conductive contact pads from beyond the outside surface of the first substrate.

In an embodiment, the microelectronic assembly comprises one or more electrically conductive structures disposed within one or more of the one or more secondary openings, the one or more primary openings, and the one or more tertiary openings, and electrically coupled to the one or more electrically conductive contact pads.

In another implementation, a microelectronic assembly comprises a first substrate having a bonding surface with a planarized topography, having a first plurality of electrically conductive features at the bonding surface of the first substrate, and a second substrate having a bonding surface with a planarized topography, bonded to the bonding surface

5

of the first substrate. A second plurality of electrically conductive features is disposed at the bonding surface of the second substrate and bonded to the first plurality of electrically conductive features while misaligned to the first plurality of electrically conductive features by a first extent. One or more electrically conductive contact pads is disposed within an insulating layer of the second substrate and below the bonding surface of the second substrate. The one or more electrically conductive contact pads is disposed either outside a perimeter of the first plurality of electrically conductive features and the second plurality of electrically conductive features or in one or more designated areas. One or more secondary openings in the insulating layer of the second substrate is aligned to the one or more electrically conductive contact pads, and the one or more secondary openings extend from the bonding surface of the second substrate to the one or more electrically conductive contact pads. One or more primary openings in an insulating layer of the first substrate are misaligned to the one or more secondary openings by the first extent, and the one or more primary openings extend to the one or more secondary openings and provide access to the one or more electrically conductive contact pads.

In additional implementations, a microelectronic assembly comprises a first substrate having a hybrid bonding surface comprising an insulator material with one or more metallic features therein, and a planarized topography, and a second substrate having a hybrid bonding surface comprising an insulator material with one or more metallic features embedded therein, having a planarized topography, and bonded to the hybrid bonding surface of the first substrate. The one or more metallic features of the second substrate are bonded to the one or more metallic features of the first substrate. A cavity is disposed at a bond joint between the first substrate and the second substrate where the hybrid bonding surface of the first substrate and the hybrid bonding surface of the second substrate make contact. The cavity is formed at least by a recess in the hybrid bonding layer of the first substrate. A continuous seal is formed by the one or more metallic features of the first substrate and the one or more metallic features of the second substrate, and is disposed around a periphery of the cavity at the bond joint.

Various implementations and arrangements are discussed with reference to electrical and electronics components and varied carriers. While specific components (i.e., dies, wafers, integrated circuit (IC) chip dies, substrates, etc.) are mentioned, this is not intended to be limiting, and is for ease of discussion and illustrative convenience. The techniques and devices discussed with reference to a wafer, die, substrate, or the like, are applicable to any type or number of electrical components, circuits (e.g., integrated circuits (IC), mixed circuits, ASICs, memory devices, processors, etc.), groups of components, packaged components, structures (e.g., wafers, panels, boards, PCBs, etc.), and the like, that may be coupled to interface with each other, with external circuits, systems, carriers, and the like. Each of these different components, circuits, groups, packages, structures, and the like, can be generically referred to as a "microelectronic component." For simplicity, unless otherwise specified, components being bonded to another component will be referred to herein as a "die."

This summary is not intended to give a full description. Implementations are explained in more detail below using a plurality of examples. Although various implementations and examples are discussed here and below, further imple-

6

mentations and examples may be possible by combining the features and elements of individual implementations and examples.

DETAILED DESCRIPTION

Overview

Patterned metal and inorganic dielectric material, such as silicon oxide, layers are frequently provided on a die, wafer, or other substrate (hereinafter "die") as a hybrid bonding, or DBI®, surface layer. The dielectric layer is typically highly planar (usually to nm-level roughness) with the metal layer (e.g., embedded conductive features) at or recessed just below the oxide surface. The amount of recess below the oxide is typically determined by a dimensional tolerance, specification, or physical limitation. The hybrid surface is often prepared for bonding with another die, wafer, or other substrate using a chemical-mechanical polishing (CMP) step.

The devices and techniques disclosed herein allow further etching of the oxide (or other insulating layer) after the CMP step to form an opening, cavity, or recess in or through the bonding layer. The openings or recesses allow for features or structures to be formed and accessed, such as wirebond pads, testing pads, die cavities, and so forth. Forming such cavities may mean that additional openings need to be etched into the hybrid bonding surface layer after the CMP step, but prior to bonding. Such cavities may be used to create access ports for sensor applications (e.g., DNA analysis), physical and electrical access (e.g. for testing, wire-bonding, electrical connection, etc.), low-impedance, low-loss connections, air bridges, and the like.

The techniques described herein are useful to form shallower cavities, recesses or openings in the individual dies or wafers prior to bonding, thereby avoiding having to etch a deeper single cavity in both dies or wafers after bonding. This process also results in a cleaner bond pad surface and finer pitch capability than would be possible with oxide etching after bonding. In addition, the process also allows for the formation of cavities with different dimensions in the top and bottom wafer or die. For example, a trench opening may be formed in the bottom die and a spot opening may be formed in the top die, or vice versa. Alternatively, an opening formed in the bottom die can be larger than an opening formed in the top die, or vice versa (see FIGS. 29-32).

As will be appreciated, this cavity etching technique can improve metal seal ring or electrical contact for a microelectromechanical systems (MEMS) or other sensor applications, as well. The techniques described herein also improve bonding through a process of street etching before die singulation. This improves dicing by reducing chipping and improves transistor reliability in the field.

FIGS. 1-33 illustrate representative devices and processes for forming and preparing various microelectronic components (such as dies 102, for example) for bonding, such as for direct bonding without adhesive. The processes include providing a bonding surface (such as bonding surface 108, for example) on the microelectronic components, or two bonding surfaces in some examples (not shown), providing conductive interconnect features or structures embedded into the bonding surfaces, repairing or mitigating erosion, dishing, and the like in the bonding surfaces due to processing or defects, pre-etching one or more openings in the bonding surfaces, forming microelectronic assemblies by

directly bonding the microelectronic components at the bonding surfaces, and so forth.

The order in which the processes are described is not intended to be construed as limiting, and any number of the described process blocks in the processes can be combined in any order to implement the processes, or alternate processes. Additionally, individual blocks may be deleted from any of the processes without departing from the spirit and scope of the subject matter described herein. Furthermore, the processes can be implemented in any suitable hardware, software, firmware, or a combination thereof, without departing from the scope of the subject matter described herein. In alternate implementations, other techniques may be included in the processes in various combinations and remain within the scope of the disclosure.

Referring to FIGS. 1 and 2, and processes 100 and 200 respectively, a representative “die” 102 may be formed using various techniques, to include a base substrate 104 and an insulating or dielectric layer 106. The base substrate 104 may be comprised of silicon, germanium, glass, quartz, a dielectric surface, direct or indirect gap semiconductor materials or layers or another suitable material, plus multiple layers of metal wiring in insulating dielectrics that are commonly referred to as the Back-End-of-Line (BEOL) layers. The insulating layer 106 is deposited or formed over the substrate 104, and may be comprised of an inorganic dielectric material layer such as oxide, nitride, oxynitride, oxycarbide, carbides, carbonitrides, diamond, diamond like materials, glasses, ceramics, glass-ceramics, and the like. The insulating layer 106 can be the last layer of BEOL, or an additional layer deposited specifically for hybrid bonding, for instance.

A bonding surface 108 of the die 102 can include conductive features 110, such as interconnect structures for example, embedded into the insulating layer 106 and arranged so that the conductive features 110 from respective bonding surfaces 108 of opposing dies 102 can be mated and joined during bonding, if desired. The joined conductive features 110 can form continuous conductive interconnects (for signals, power, etc.) between stacked dies 102.

Damascene processes (or the like) may be used to form the embedded conductive features 110 in the insulating layer 106. The conductive features 110 may be comprised of metals (e.g., copper, etc.) or other conductive materials, or combinations of materials, and include structures, traces, pads, patterns, and so forth. The conductive features 110 may be included in the insulating layer 106 to provide an electrical and/or thermal path or may instead be configured to balance out the metallization of the bonding surface 108, through the use of additional pads or so-called dummy pads, traces, patterns or the like. After the conductive features 110 are formed, the exposed surface of the die 102, including the insulating layer 106 and the conductive features 110 can be planarized (e.g., via CMP) to form a flat bonding surface 108.

As shown in FIGS. 1 and 2, at block A of processes 100 and 200, one or more bonding surfaces 108 of a die 102, including embedded interconnect structures 110, can be planarized (using chemical-mechanical polishing (CMP), or the like) to prepare the surface(s) 108 for bonding. Forming the bonding surface 108 includes finishing the surface 108 to meet dielectric roughness specifications and metallic layer (e.g., copper, etc.) recess specifications, to prepare the surface 108 for direct bonding. In other words, the bonding surface 108 is formed to be as flat and smooth as possible, with very minimal surface topography variance. Various conventional processes, such as chemical mechanical pol-

ishing (CMP) may be used to achieve the low surface roughness. This process provides the flat, smooth surface 108 that results in a reliable bond between dies 102.

In some cases, as shown in FIG. 1, process 100, block A, the exposed surface of the conductive features 110 may be intentionally recessed relative to the bonding surface 108, for example to a depth “d1,” to allow for material expansion, particularly during heated annealing, if it is to be performed. In other cases, as shown in FIG. 2, process 200, block A, the exposed surface of the conductive features 110 may be formed to exceed the recess specification, and may protrude above the bonding surface 108 to a height “d2,” to allow for oxidation of the conductive features 110 during later processing. This may be accomplished by selective etching of the dielectric layer 106, for example.

During polishing or other process steps (such as forming an opening in the bonding layer 108 after the polishing step), the conductive features 110 at the bonding layer 108 may change (e.g., become oxidized) and become out of the desired specification. This is illustrated at FIGS. 1 and 2, processes 100 and 200, block B, where the oxidation is shown at 112.

The conductive features 110 may be adjusted, as shown at FIGS. 1 and 2, processes 100 and 200, block C, to improve the subsequent bonding and electrical connection. The adjustment can be achieved by a selective wet etch of the conductive features 110, for instance with a chemical etch that selectively dissolves the metal oxide (e.g., copper oxide) formed on the conductive features 110, or by a touch-up CMP process, or the like. After the adjustment, the recess of the conductive features 110 relative to the bonding surface 108 is within tolerance of the specification (e.g., “d3” and “d4”).

Example Embodiments

Referring to FIG. 3, a microelectronic assembly 300 is shown, comprising at least two directly bonded (e.g., without adhesive) dies 102 and 302. Alternately, the microelectronic component 302 may comprise a wafer, or other substrate, or the like. However, the microelectronic component 302 includes one or more contact pads 306 that are disposed outside of the die 102 perimeter.

Embedded electrically conductive features 110 (e.g., hybrid bonding pads, etc.) of both dies 102 and 302 extend to the bonding surfaces 108 of the dies 102 and 302 and are bonded together, forming bonded interconnect structures 304. In an embodiment, the first die 102 may be formed as described above at processes 100 or 200.

Contact pads 306 (i.e., exposed non-hybrid conductive pads) of die 302 are exposed through the bonding surface 108 of the die 302 due to the openings 308 in the insulating layer 106 of the die 302. As discussed above, contact pads 306 may be used for testing, wirebonding, other electrical connection, and so forth regarding the die 302. In some embodiments, a conductive coating 310, comprising nickel, gold or other metals, for example, may be disposed over the contact pads 306 to protect the pads 306 from oxidation, corrosion, or the like.

In an implementation, the second die 302 may be formed as described in process 400, including forming the openings 308 prior to bonding the second die 302 to the first die 102. Alternately, in some embodiments with the structure 300, the openings 308 may be formed in the second die 302 after bonding to the first die 102.

Referring to FIG. 4, process 400 (see also FIG. 33 for a text flow diagram to complement the graphical flow diagram

of FIGS. 4 and 5), at block A, the die 302 is formed and prepared for direct bonding, including forming the insulating layer 106 on the base substrate 104, forming one or more conductive features 110 in the insulating layer 106, and forming the highly planar bonding layer 108 as described above. In an implementation, the process 400 includes forming buried contact pads 306 within the insulating layer 106.

At block B, the process 400 includes forming a patterned resist layer 402 over the bonding surface 108, with patterned gaps in the resist layer 402 aligned over the buried contact pads 306. Optionally, the patterned resist layer 402 may include gaps over the dicing street, if desired. At block C, the process 400 includes etching the insulating layer 106 through the gaps in the resist layer 402, forming the openings 308 to expose the buried contact pads 306.

Referring to FIG. 5, at block D, the process 400 includes optionally covering the uncovered contact pads 306 with a protective metallic layer 310 (e.g., nickel, gold, silver, solder, etc.) if desired to protect the pads 306. The protective layer 310 may be applied using electroless plating, for example, or another technique if desired, such as immersion plating or the like.

At block E, the process 400 includes removing the resist 402 from the bonding surface 108. The resist 402 may be removed using a wet strip followed by oxygen plasma ashing, for example. In some cases, the resist 402 removal steps may cause the surfaces of the conductive features 110 to oxidize (forming copper oxide 404, for example). Depending on the amount of oxidation 404, the surfaces of the conductive features 110 may protrude above the bonding surface 108, as shown in FIG. 5.

At block F, the process 400 includes removing the oxidation 404 from the conductive features 110. For example, the oxidation 404 may be about 1-100 nm thick. Alternately, the oxidation may be thicker in some cases. In any case, controlling the rate of material removal, including the rate of removing oxidation 404, as well as the rate of removing some of the metal (e.g., copper, etc.) of the conductive structures 110 to adhere to recess specifications, can be of importance. For instance, removing the correct amount of material can avoid having to refabricate the die 302, including reforming the insulating layer 106 and/or the conductive structures 110. Unfortunately, it can be easy to over-etch material using many of the commonly used techniques.

In various embodiments, controlled, selective etching techniques are used to remove the oxidation 404, which may include etching the oxide 404 with a dilute formulary, such as a 1:20 ratio of sulfuric acid or sulfonic acid and water. Alternately, a light CMP with a slurry formulated for polishing a barrier layer (such as a barrier layer deposited into a cavity of an insulating layer 106 prior to copper deposition during a Damascene process, for example) can be used to selectively remove the oxidation 404. However, the use of chemical etching may be more easily controlled for nanometer scale material removal. The specified recess for the conductive features 110 may be achieved as part of the oxidation 404 removal step with controlled material removal.

Referring to FIGS. 5A-5C, as alternative steps to the process 400, or in addition to the process steps in blocks D through F, one or more optional layers may be deposited over the exposed contact pads 306 (and the protective metallic layer 310), which may include depositing the layer(s) over the bonding surface 108 of the die 302. For instance in some cases, electrochemical enhancement of etching on some conductive features 110 may be experi-

enced, due to bimetal effects and photovoltaic effects for conductive features 110 coupled to circuits or exposed bond pads. These effects can result in corrosion of the contact pads 306.

In these cases, (referring to FIG. 5A at block G) a layer 502 of approximately 10-200 nm of oxide and/or barrier metal can be deposited over the bonding surface 108 and the contact pads 306 (including the protective metallic layer 310) of the die 302. As shown at block H, CMP (or the like) can be used to re-expose the conductive features 110 on the bonding surface 108. The layer 502 remains on the contact pads 306. At block I, a first die or wafer 102 is direct bonded to the second die 302, without adhesive.

Referring to FIG. 5B, at block J, after the second die 302 is bonded to the first die or wafer 102, the top base layer 104 of the first die or wafer 102 can be thinned. At block K, a patterned resist layer 402 is applied to the top surface of the die 102, with gaps in the resist at locations where it is desired to remove the base layer 104 (for instance, over the openings 602 as well as other locations, as desired). At block L, openings are etched in the base layer 104 of the wafer or die 102 to access the contact pads 306. The openings in the base layer 104 of the die 102 can be formed to be the same size as the openings in the insulating layer 106 of either or both of the die 102 and the die 302, or it can be larger or smaller than the mating openings in the die 102 and/or the die 302.

Referring to FIG. 5C, at block M, with openings created in the base layer 104 of the wafer or die 102, the thin layer 502 over the contact pad 306 can be removed with a brief blanket dry etch, or like technique. Then the photo resist 402 can be removed as shown at block N. The resist 402 may be removed using a wet strip, for example. The resulting microelectronic assembly 500 is shown at block N.

Referring to FIG. 6, a microelectronic assembly 600 is shown, comprising at least two directly bonded (e.g., without adhesive) dies 102 and 302. Alternately, the microelectronic components 102 and 302 may comprise two dies, two wafers, one die and one wafer, or other substrates, or the like. However, the microelectronic component 302 includes one or more contact pads 306 that are disposed inside of the die 102 perimeter.

Embedded conductive features 110 (e.g., hybrid bonding pads, etc.) of both dies 102 and 302 extend to the bonding surfaces 108 of the dies 102 and 302 and are bonded together, forming bonded interconnect structures 304. In an embodiment, the first die 102 may be formed as described at process 700.

Referring to FIG. 7, and process 700, at block A, the die 102 is formed and prepared for direct bonding, including forming the insulating layer 106 on the base substrate 104, forming one or more conductive features 110 in the insulating layer 106, and forming the highly planar bonding layer 108 (using CMP, for instance) as described above. In various implementations, block A may be performed on a wafer, and the steps of process 700 may be performed prior to singulation (if desired).

At block B, the process 700 includes forming a patterned resist layer 402 over the bonding surface 108, with patterned gaps in the resist layer 402 aligned over a portion of the insulating layer 106 corresponding to locations for the buried contact pads 306 in the die 302, once the dies 102 and 302 are aligned and bonded. Optionally, the opening in die 102 can be formed to be the same size as the opening in die 302, or it can be larger or smaller than the mating opening in die 302. Optionally, the patterned resist layer 402 may include gaps over the dicing street, if desired. At block C, the process 700 includes etching the insulating layer 106 (and

11

some of the base substrate **104**, if desired) through the gaps in the resist layer **402**, forming openings **702** in the insulating layer **106**.

Referring to FIG. 8, at block D, the process **700** includes removing the resist **402** from the bonding surface **108**. The resist **402** may be removed using a wet strip followed by oxygen plasma ashing, for example. In some cases, the resist **402** removal steps may cause the surfaces of the conductive features **110** to oxidize (forming copper oxide **404**, for example). Depending on the amount of oxidation **404**, the surfaces of the conductive features **110** may protrude above the bonding surface **108**, as shown in FIG. 8.

At block E, the process **700** includes removing the oxidation **404** from the conductive features **110** using a CMP process with a slurry formulated for polishing a barrier layer, or the like, and/or a wet etch process as described above, while controlling the removal rate of material on the conductive features **110**. For example, selective etching techniques are used to remove the oxidation **404**, which may include etching the oxide **404** with a dilute formulary, such as a 1:20 ratio of sulfuric acid or sulfonic acid and water. The specified recess for the conductive features **110** may be achieved as part of the oxidation **404** removal step with controlled material removal.

In some cases, electrochemical enhancement of etching on some conductive features **110** may be experienced, due to bimetal effects and photovoltaic effects, for example. In these cases, 10 nm to 200 nm of oxide and/or barrier metal can be deposited over the bonding surface **108** and contact pad **306**. CMP (or the like) can be used to re-expose the conductive features **110** on the bonding surface **108**.

In an implementation, the second die **302** may be formed as described in process **900**, including forming openings **308** in the insulating layer **106** of the die **302** prior to bonding the second die **302** to the first die **102**.

Referring to FIG. 9, process **900**, at block A, the die **302** is formed and prepared for direct bonding, including forming the insulating layer **106** on the base substrate **104**, forming one or more conductive features **110** in the insulating layer **106**, and forming the highly planar bonding layer **108** as described above. In an implementation, the process **900** includes forming buried contact pads **306** within the insulating layer **106**. In various implementations, block A may be performed on a wafer, and the steps of process **900** may be performed prior to singulation (if desired).

At block B, the process **900** includes forming a patterned resist layer **402** over the bonding surface **108**, with patterned gaps in the resist layer **402** aligned over the buried contact pads **306**. Optionally, the patterned resist layer **402** may include gaps over the dicing street, if desired. At block C, the process **900** includes etching the insulating layer **106** through the gaps in the resist layer **402**, forming the openings **308** to expose the buried contact pads **306**.

Referring to FIG. 10, at block D, the process **900** includes optionally plating the uncovered contact pads **306** with a protective metallic layer **310** (e.g., nickel, gold, silver, solder, etc.) if desired to protect the pads **306**. The protective layer **310** may be applied using electroless plating, for example, or another technique if desired, such as immersion plating or the like.

At block E, the process **900** includes removing the resist **402** from the bonding surface **108**. The resist **402** may be removed using a wet strip followed by oxygen plasma ashing, for example. In some cases, the resist **402** removal steps may cause the surfaces of the conductive features **110** to oxidize (forming copper oxide **404**, for example). Depending on the amount of oxidation **404**, the surfaces of

12

the conductive features **110** may protrude above the bonding surface **108**, as shown in FIG. 10.

At block F, the process **900** includes removing the oxidation **404** from the conductive features **110** using a CMP process and/or a wet etch process as described above, while controlling the removal rate of material on the conductive features **110**. For example, selective etching techniques are used to remove the oxidation **404**, which may include etching the oxide **404** with a dilute formulary, such as a 1:20 ratio of sulfuric acid (or sulfonic acid) and water, or a light CMP with a slurry formulated for polishing a barrier layer, or the like. The specified recess for the conductive features **110** may be achieved as part of the oxidation **404** removal step with controlled material removal.

In some cases, electrochemical enhancement of etching on some conductive features **110** may be experienced, due to bimetal effects and photovoltaic effects, for example. In these cases, approximately 10 nm to 200 nm of oxide and/or barrier metal can be deposited over the bonding surface **108** and over the contact pads **306** (not shown). CMP (or the like) can be used to re-expose the conductive features **110** on the bonding surface **108**. After the second die **302** is bonded to the first die **102**, the contact pads **306** can be accessed through openings created in the first die **102** and the pads **306** can be uncovered with a dry etch.

Referring back to FIG. 6, direct bonding the prepared dies **102** and **302** at the bonding surfaces **108** of each yields the microelectronic assembly **600**. As shown, the opening **308** in the die **302** combines with the opening **702** in the die **102** to form a cavity **602** above the contact pads **306**. In some embodiments, the base substrate **104** (which may comprise silicon, for example) of the first die **102** may be etched above the cavity **602** and the contact pads **306** to provide access to the contact pads **306** from outside the microelectronic assembly **600**, if desired. In such embodiments, etching through the base layer **104** alone is preferred to etching through the base layer **104** and the insulating layer **106** of the die **102** while the first die **102** is bonded to the second die **302**. Etching through the single layer **104** is much easier with the insulating layer **106** already removed, which improves the fabrication process. Further, this can result in a finer pitch opening over the pads **306** than would be possible when etching through both the base layer **104** and the insulating layer **106** of the die **102**. In addition, this allows openings of different sizes and shapes within each of the two dies **102** and **302** bonded together, if desired.

Referring to FIG. 11, a process **1100** for forming a microelectronic assembly **1200** is shown, comprising at least two directly bonded (e.g., without adhesive) dies **102** and **302**. Alternately, the microelectronic component **302** may comprise a wafer, or other substrate, or the like. However, the microelectronic component **302** includes one or more contact pads **306** that are disposed inside of the die **102** perimeter.

Embedded conductive features **110** (e.g., hybrid bonding pads, etc.) of both dies **102** and **302** extend to the bonding surfaces **108** of the dies **102** and **302** and are bonded together, forming bonded interconnect structures **304**. In an embodiment, the first die **102** may be formed as described above at process **700** and the second die **302** may be formed as described above at process **900**. Accordingly, the process **1100** begins with a microelectronic assembly **600**, or the like, where the first die **102** may have a footprint that is smaller, larger, or substantially the same size as the second die **302**. In some cases, the first **102** and second **302** dies may be substantially aligned, or they may be slightly misaligned, but within hybrid bonding tolerances.

13

Referring to FIG. 11, at block A, an example microelectronic assembly is shown, such as the microelectronic assembly 600 discussed previously. At block B, the process 1100 includes thinning the top base wafer 104 of the first die 102 as desired. At block C, the process 1100 includes forming a patterned resist layer 1102 over the base layer 104 of the first die 102, with patterned gaps in the resist layer 1102 aligned over portions of the base layer 104 corresponding to locations where it is desired to remove portions of the base layer 104, including for example, over the cavities 602 and the buried contact pads 306 in the die 302, over non-active areas of the die 102, as well as other areas of the die 102 as desired.

Referring to FIG. 12, at block D, the process 1100 includes etching the material (e.g., silicon, etc.) of the base layer 104 of the die 102 to reveal areas below, including the contact pads 306 (or other structure(s) in the cavity 602) in the second die 302. As shown, active base material (e.g., silicon) islands 1202 may be left on the top surface of the die 102 in active areas of the die 102. At block E, the process 1100 includes removing the resist 1102 from the base layer 104. The resist 1102 may be removed using a wet strip followed by oxygen plasma ashing, for example. This yields the microelectronic element 1200, as shown. At block F, the process 1100 may include an optional step where the base material islands 1202 may be electrically insulated and/or EMI shielded, or the like. In the case of an image sensor, laser, light emitting diode (LED), etc., an anti-reflection coating 1204, or the like, may be applied to the islands 1202.

Referring to FIG. 13, a process 1300 for forming a microelectronic assembly 1302 is shown, comprising at least two directly bonded (e.g., without adhesive) dies 102 and 302. Alternately, the microelectronic component 302 may comprise a wafer, or other substrate, or the like. However, the microelectronic component 302 includes one or more contact pads 306 that are disposed inside of the die 102 perimeter.

Embedded conductive features 110 (e.g., hybrid bonding pads, etc.) of both dies 102 and 302 extend to the bonding surfaces 108 of the dies 102 and 302 and are bonded together, forming bonded interconnect structures 304. In an embodiment, the first die 102 may be formed as described above at process 700 and the second die 302 may be formed as described above at process 900. Accordingly, the process 1300 begins with a microelectronic assembly 600, or the like, where the first die 102 may have a footprint that is smaller, larger, or substantially the same size as the second die 302. In some cases, the first 102 and second 302 dies may be substantially aligned, or they may be slightly misaligned, but within hybrid bonding tolerances.

At block A, an example microelectronic assembly is shown, such as the microelectronic assembly 600 discussed previously. At block B, the process 1300 includes forming a patterned resist layer 1102 over the base layer 104 of the first die 102, with patterned gaps in the resist layer 1102 aligned over a portion of the base layer 104 corresponding to locations for the cavities 602 and the buried contact pads 306 in the die 302. Additionally, as shown at block B, resist may be deposited at other locations, such as at predetermined portions of the bonding layer 108 of the die 302, or the like. Resist is placed at the additional locations to prevent etching of material at these locations.

At block C, the process 1300 includes etching the material (e.g., silicon, etc.) of the base layer 104 of the die 102 to reveal the contact pads 306 in the second die 302. At block D, the process 1300 includes removing the resist 1102 from the base layer 104. The resist 1102 may be removed using a

14

wet strip followed by oxygen plasma ashing, for example. This yields the microelectronic element 1302, as shown.

Additional Embodiments

Referring to FIG. 14, a process 1400 for forming a microelectronic assembly 1500 is shown. In various embodiments, the assembly 1500 is formed from a prepared die 102, having metallic seal rings 1402, disposed within the die 102 (e.g., embedded within the insulating layer 106, and extending into the base substrate 104 in some cases) as desired. In some examples, the hermeticity of the finished assembly 1500 can be based on the dimensions of the seal rings 1402.

The die 102 is prepared as described in processes 100 or 200, for example, using a Damascene process (or the like) to form the seal rings 1402. The assembly 1500 also includes one or more cavities 1404 that can be terminated inside the insulating layer 106, terminated at the interface of insulating layer 106 and the base layer 104, or extend through the insulating layer 106 and may extend at least partly through the base layer 104 of the die 102 as well. Accordingly, the assembly 1500 is well suited to applications with cavities, such as MEMS, sensors, and the like. The assembly 1500 may be formed and prepared for direct bonding to another die, wafer, substrate, or the like, or another assembly, such as another assembly 1500 as described herein.

At block A, the process 1400 includes planarizing the hybrid bonding surface 108 of the first die 102 as desired, using a CMP process for example. At block B, the process 1400 includes forming a patterned resist layer 402 over the bonding surface 108 of the first die 102, with patterned gaps in the resist layer 402 aligned over a portion of the insulating layer 104 corresponding to locations for cavities 1404 to be formed. For instance, the cavities 1404 may be formed inside the perimeter of the seal rings 1402.

At block C, the process 1400 includes etching the materials (e.g., oxide, silicon, etc.) of the insulating layer 106 (partially or completely) and optionally partially into the base layer 104 of the die 102 to a desired depth to form one or more cavities 1404.

Referring to FIG. 15, at block D, the process 1400 includes removing the resist 402 from the bonding surface 108. The resist 402 may be removed using a wet strip followed by oxygen plasma ashing, for example. In some cases, the resist 402 removal steps may cause the surfaces of the seal rings 1402 to oxidize (forming copper oxide 404, for example). Depending on the amount of oxidation 404, the surfaces of the seal rings 1402 may protrude above the bonding surface 108, as shown in FIG. 15.

At block E, the process 1400 includes removing the oxidation 404 from the seal rings 1402 using a CMP process and/or a wet etch process as described above, while controlling the removal rate of material on the seal rings 1402. For example, selective etching techniques are used to remove the oxidation 404, which may include etching the oxide 404 with a dilute formulary, such as a 1:20 ratio of sulfuric acid and water, a touch-up CMP with a slurry formulated for polishing a barrier layer, or the like. A specified recess for the seal rings 1402 (if applicable) may be achieved as part of the oxidation 404 removal step with controlled material removal.

In some cases, electrochemical enhancement of etching on some seal rings 1402 may be experienced, due to bimetal effects and photovoltaic effects, for example. In these cases, approximately 10-200 nm of oxide and/or barrier metal can be deposited over the bonding surface 108. CMP (or the

15

like), with a barrier layer slurry, for example, can be used to touch polish and re-expose the seal rings **1402** on the bonding surface **108**. Care must be taken to avoid rounding the edges (e.g., eroding) of the insulating layer **106** as it meets the seal rings **1402**. The result of the process **1400** is the microelectronic element **1500**, as shown, which is ready for direct bonding.

Referring to FIGS. **16** and **17**, a process **1600** for street etching prior to die **102** singulation is shown. In various embodiments, a wafer **1602** can be formed and prepared as described in processes **100** or **200**, for example, using a Damascene process (or the like) to form one or more conductive features **110** (or seal rings **1402**, etc.). At block A, the process **1600** includes planarizing the hybrid bonding surface **108** as desired, using a CMP process for example. At block B, the process **1600** includes forming a patterned resist layer **402** over the bonding surface **108** of the dies **102** of the wafer **1602**, with patterned gaps in the resist layer **402** aligned over a portion of the insulating layer **106** corresponding to a dicing street **1604** for die **102** singulation. For instance, the dicing street **1604** is disposed between formed dies **102** of the wafer **1602**.

At block C, the process **1600** includes etching the materials (e.g., oxide, silicon, etc.) of the insulating layer **104** and the base layer **104** of the die **102** to a desired depth to form the dicing street **1604** channel or recess **1606**.

Referring to FIG. **17**, at block D, the process **1600** includes removing the resist **402** from the bonding surface **108**. The resist **402** may be removed using a wet strip followed by oxygen plasma ashing, for example. In some cases, the resist **402** removal steps may cause the surfaces of the conductive features **110** (or seal rings **1402**, for example) to oxidize (forming copper oxide **404**, for example). Depending on the amount of oxidation **404**, the surfaces of the conductive features **110** may protrude above the bonding surface **108**, as shown in FIG. **17**.

At block E, the process **1600** includes removing the oxidation **404** from the conductive features **110** using a CMP process and/or a wet etch process as described above, while controlling the removal rate of material on the conductive features **110**. For example, selective etching techniques are used to remove the oxidation **404**, which may include etching the oxide **404** with a dilute formulary, such as a 1:20 ratio of sulfuric acid and water, or by light CMP with a slurry formulated for polishing a barrier layer, or the like. A specified recess for the conductive features **110** (if applicable) may be achieved as part of the oxidation **404** removal step with controlled material removal.

In some cases, electrochemical enhancement of etching on some conductive features **110** may be experienced, due to bimetal effects and photovoltaic effects, for example. In these cases, approximately 10 nm-200 nm of oxide and/or barrier metal can be deposited over the bonding surface **108**. CMP (or the like), with a barrier layer slurry, for example, can be used to touch polish and re-expose the conductive features **110** on the bonding surface **108**. Care must be taken to avoid rounding the edges (e.g., eroding) of the insulating layer **106** as it meets the conductive features **110**.

At block F, as an option, a protective coating **1702** may be applied to the bonding surface, as well as within the dicing street recess **1606**, if desired. The protective coating can protect the bonding surface **108** during singulation into individual dies **102**, the singulation occurring within the dicing street recess **1606**.

In various embodiments, wafers (such as wafers **1602**), dies (such as dies **102** and **302**), and other substrates and so forth having a hybrid bonding surface **108** may be touch

16

polished after long-term storage results in some oxidation of metallic features, such as conductive features **110**, seal rings **1402**, and the like. The touch polish may include using a CMP process and/or a wet etch process as described above, while controlling the removal rate of material on the metallic features. For example, selective etching techniques are used to remove the oxidation, which may include etching the oxide with a dilute formulary, such as a 1:20 ratio of sulfuric acid and water, touch polish with a slurry formulated for polishing a barrier layer, or the like. A specified recess for the metallic features (if applicable) may be restored as part of the oxidation removal step with controlled material removal.

In some cases, electrochemical enhancement of etching on some metallic features may be experienced, due to bimetal effects and photovoltaic effects, for example. In these cases, approximately 10 nm-200 nm of oxide and/or barrier metal can be deposited over the bonding surface **108**. CMP (or the like), with a barrier layer slurry, for example, can be used to touch polish and re-expose the metallic features on the bonding surface **108**. Care must be taken to avoid rounding the edges (e.g., eroding) of the insulating layer **106** as it meets the metallic features. After touch polishing, the wafers, dies, substrates, etc. are prepared for direct bonding.

Additional Example Assemblies

Referring to FIGS. **18-32**, in various embodiments, various microelectronic assemblies may be formed, based on or related to the assemblies **300**, **600**, **1200**, **1302**, **1500**, and so forth. In other words, the various microelectronic assemblies comprise assemblies of two or more direct bonded dies **102** and/or **302**, wafers **1602**, substrates, and the like, in various combinations.

Referring to FIG. **18**, at block A, a microelectronic assembly **1800** is shown, comprising an assembly **600** with one or more contact pads **306** that are disposed inside of the die **102** perimeter. The assembly **1800** includes openings etched through the base layer **104** of the die **102** to reveal the contact pads **306** in the second die **302** and the cavities **602**, as shown in FIG. **12**, block E.

As shown in FIG. **18**, in some cases, the first **102** and second **302** dies (and the respective conductive features **110**) may be substantially aligned, or they may be slightly misaligned, but within hybrid bonding tolerances. The openings **702** and **308** in the first **102** and second **302** dies may also be somewhat misaligned, but still form the cavity **602** and still provide sufficient access to the contact pads **306**.

Optionally, as shown in block B, one or more conductive interconnect structures **1802** may be electrically coupled to the contact pads **306**, to provide electrical access to the contact pads **306** from the top of the assembly **1800**. For instance, the conductive interconnect structures **1802** may protrude above the top of the base layer **104** of the die **102** in some cases.

In an embodiment, as shown at FIG. **19**, the first die **102** of the assembly **1800** may have a smaller footprint than the second die **302**. In another embodiment, as also shown at FIG. **19**, not all of the cavities **602** may include a conductive interconnect structure **1802**. Access to the contact pad **306** may be from inside the cavity **602**, or the like.

In an embodiment, as shown at FIG. **20**, a microelectronic assembly **2000** may comprise an assembly **300** where the first die **102** of the assembly **2000** has a smaller footprint than the second die **302**, and where the non-hybrid contact pads **306** of the second die **302** are disposed outside of the

perimeter of the first die **102**. In an alternate implementation, the assembly **200** includes one or more conductive interconnect structures **1802** electrically coupled to the contact pads **306**, to provide electrical access to the contact pads **306** from the top of the assembly **2000**.

In an embodiment, as shown at FIG. **21**, a microelectronic assembly **2100** may comprise an assembly **1200** where the first die **102** of the assembly **2000** has a smaller, larger, or substantially same size footprint than the second die **302**, and where the non-hybrid peripheral contact pads **306** of the second die **302** are disposed inside of the perimeter of the first die **102**. Further, the first die **102** includes islands **1202** formed by etching off the top base layer **104** of the die **102** in non-active areas of the die **102**. For instance, active base material (e.g., silicon) islands **1202** may be left on the top surface of the die **102** in active areas of the die **102**. In some embodiments, the base material islands **1202** may be electrically insulated and/or EMI shielded, or the like. In the case of an image sensor, laser, light emitting diode (LED), etc., an anti-reflection coating may be applied to the islands **1202**.

In an alternate embodiment, as shown at FIG. **22**, the assembly **2100** includes at least one metal layer **2202** on the exposed insulating layer **106** of the first die **102**. In various examples, the metal layers **2202** can provide EMI protection, heat dissipation, or other functions as desired.

In another alternate embodiment, as shown at FIG. **23**, the assembly **2100** may include deposited metal **2302** in one or more of the cavities **602**, electrically coupled to the contact pads **306** for interconnection to the pads **306**. For instance, in some cases, a terminal **2304** (such as a solder ball, or the like) may be coupled to the deposited metal **2302** for convenience in making the desired interconnections.

Referring to FIGS. **24-26**, direct bonded assemblies **2400** with cavities **1404** and seal rings **1402** are shown. In various implementations, the assemblies **2400** are formed of at least a pair of assemblies **1500**, directly bonded without adhesive (e.g., hybrid bond). In an embodiment as shown in FIG. **24**, the cavity **1404** of each of the assemblies **1500** is etched through the insulating layer **106** and partly through the base layer **104**. In various alternate embodiments, the cavity **1404** may be more or less etched into one of the assemblies **1500** than the other assembly **1500'**. See below for some example etch depth options.

In one example, as shown in FIG. **25**, the cavity **1404** is etched into the first assembly **1500**, and not etched into the second assembly **1500'**. The example etch depth shown in the drawing of FIG. **25** is through the insulating layer **106** and partially into the base layer **104**. Alternatively, the etch depth may be partially through the insulating layer **106** only, or through the insulating layer **106** but not into the base layer **104**. In alternate embodiments, the cavity **1404** may be etched into the second assembly **1500'** and not into the first assembly **1500** (see above for example etch depth options). Further, as shown in FIG. **26**, an assembly **2400** may include one or more components **2602** (e.g., components, structures, circuits, elements, etc.) mounted inside the cavity **1404**. Since the assembly **2400** includes seal rings **1402**, the cavity **1404** can be hermetically sealed, providing an ideal environment for various sensors, MEMS, or the like.

Referring to FIGS. **27-28**, in another embodiment, an assembly **2700** comprises an assembly **2400** where the second assembly **1500'** has a footprint that is larger than a footprint of the first assembly **1500**. In an example, the second assembly **1500'** includes a peripheral bond pad **2702** outside the perimeter of the first assembly **155**, which may be used for wirebond **2704**, or like electrical connections. The cavity **1404** of the assembly may include one or more

components **2602**, and the cavity **1404** may be hermetically sealed, due to the presence of seal rings **1402**. Also, the assembly **2700** may include one or more conductive features **110** for electrical interconnection within the assembly **2700**, and to outside components, for example using a through-silicon via (e.g., TSV) **2706**, or the like.

As shown in FIG. **28**, a plurality of assemblies **2700** may be stacked to form an assembly **2800**. Alternately, as shown, at least one or more of the plurality of assemblies **2700** may not include a base layer **104** on one of the assemblies **1500** and/or **1500'**. As shown, the assemblies may include seal rings **1402** as well as conductive features **110** (these arranged to electrically couple the assemblies **2700** together, with or without vias **2706**).

Referring to FIGS. **29-32**, a microelectronic assembly **2900** is shown, comprising an assembly **500** with one or more contact pads **306** that are disposed inside of the die **102** perimeter. For instance, at FIGS. **29** and **31** show a microelectronic assembly **2900** with the contact pads **306** disposed between two active areas (**2902** and **2904**) of the assembly **2900**. Thus, the contact pads **306** may be disposed at preselected areas that are not necessarily at the periphery of the dies **102** and **302**. For instance, the contact pads **306** may be disposed within a perimeter of the active area of a die **102** and **302**. The assembly **2900** includes openings etched through the base layer **104** and the insulating layer **106** of the die **102**, as well as the insulating layer **106** of the die **302** to reveal the cavities **602** and the contact pads **306** in the second die **302**.

FIG. **30** illustrates a top-down view of an assembly **2900**, or the like, showing that the contact pads **306** may be in the form of conductive strips, or other shaped structures. For example, stacked Dynamic Random Access Memory (DRAM) assemblies often have TSVs in a center strip (or like locations) of the die **302**. This arrangement may be used with other assemblies as well.

As shown at FIG. **31**, the etched openings in the base layer **104** and insulating layer **106** of the first die **102** and the insulating layer **106** of the second die **302** that form the cavities **602** for access to the contact pads **306** need not be perfectly aligned. Additionally, as shown at FIG. **32**, the etched openings in the base layer **104** and insulating layer **106** of the first die **102** and the insulating layer **106** of the second die **302** may have different sizes, shapes, footprints, and so forth. For instance, as shown at FIG. **32**, the opening **3202** in the base layer **104** of the first die **102** may comprise a first shape (e.g., a rectangle, for this example, but may comprise any geometric or irregular shape) that exposes the insulating layer(s) **106** of the first **102** and/or second dies **302**. Additionally, the openings **3204** and **3206** in the insulating layers **106** of the first **102** and second dies **302** may comprise a same or different shape (e.g., an ellipse and a rectangle, respectively, in this example, but may comprise any geometric or irregular shape) that exposes the contact pads **306** of the second die **302**.

In various embodiments, some process steps may be modified or eliminated, in comparison to the process steps described herein.

The techniques, components, and devices described herein are not limited to the illustrations of FIGS. **1-28**, and may be applied to other designs, types, arrangements, and constructions including with other electrical components without departing from the scope of the disclosure. In some cases, additional or alternative components, techniques, sequences, or processes may be used to implement the techniques described herein. Further, the components and/or

19

techniques may be arranged and/or combined in various combinations, while resulting in similar or approximately identical results.

CONCLUSION

Although the implementations of the disclosure have been described in language specific to structural features and/or methodological acts, it is to be understood that the implementations are not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as representative forms of implementing example devices and techniques.

What is claimed is:

1. A microelectronic assembly, comprising:

a first substrate having a bonding surface, the bonding surface of the first substrate having a planarized topography;

a first plurality of electrically conductive features at the bonding surface of the first substrate;

a second substrate having a bonding surface, the bonding surface of the second substrate having a planarized topography and direct hybrid bonded to the bonding surface of the first substrate;

a second plurality of electrically conductive features at the bonding surface of the second substrate bonded to the first plurality of electrically conductive features while misaligned to the first plurality of electrically conductive features by a first extent, wherein the first and second plurality of electrically conductive features together define a footprint perimeter;

one or more electrically conductive contact pads disposed within an insulating layer of the second substrate and below the bonding surface of the second substrate, the one or more electrically conductive contact pads disposed inside the footprint perimeter;

one or more secondary openings in the insulating layer of the second substrate aligned to the one or more electrically conductive contact pads, the one or more secondary openings extending from the bonding surface of the second substrate to the one or more electrically conductive contact pads; and

one or more primary openings in an insulating layer of the first substrate, misaligned to the one or more secondary openings by the first extent, the one or more primary openings extending to the one or more secondary openings and providing electrical access to the one or more electrically conductive contact pads.

2. The microelectronic assembly of claim 1, further comprising one or more tertiary openings in a base layer of the first substrate, aligned to the one or more primary openings in the insulating layer of the first substrate, the one or more tertiary openings extending from an outside surface of the first substrate to the one or more primary openings, providing electrical access to the one or more electrically conductive contact pads from beyond the outside surface of the first substrate.

3. The microelectronic assembly of claim 2, further comprising one or more electrically conductive structures disposed within one or more of the one or more secondary openings, the one or more primary openings, and the one or more tertiary openings, and electrically coupled to the one or more electrically conductive contact pads.

4. The microelectronic assembly of claim 1, wherein a footprint of the first substrate is smaller than a footprint of the second substrate.

20

5. A microelectronic assembly, comprising:

a first substrate in a first component, the first component having a bonding surface with a planarized topography; a first plurality of electrically conductive features at the bonding surface of the first component;

a second substrate in a second component, the second component having a bonding surface, wherein the bonding surface of the second component has a planarized topography and is direct hybrid bonded to the bonding surface of the first component;

a second plurality of electrically conductive features at the bonding surface of the second component and bonded to the first plurality of electrically conductive features;

one or more electrically conductive contact pads disposed within an insulating layer of the second component and below the bonding surface of the second component, the one or more electrically conductive contact pads disposed in an area different from the first plurality of electrically conductive features and the second plurality of electrically conductive features and outside a perimeter of the first substrate; and

one or more secondary openings in the insulating layer of the second component providing electrical access to the one or more electrically conductive contact pads.

6. The microelectronic assembly of claim 5, wherein the first substrate comprises a direct band gap semiconductor material.

7. The microelectronic assembly of claim 6, wherein the first substrate comprises gallium.

8. The microelectronic assembly of claim 7, wherein the second substrate comprises silicon.

9. The microelectronic assembly of claim 7, wherein the first substrate comprises arsenide.

10. The microelectronic assembly of claim 5, wherein the first substrate comprises a base semiconductor material that has been etched to leave at least one island in an active area of the first component.

11. The microelectronic assembly of claim 10, further comprising a metal layer overlying at least a portion of the first component on a side opposite the bonding surface of the first component.

12. A microelectronic assembly, comprising:

a first substrate in a first component, the first component having a bonding surface with a planarized topography; a first plurality of electrically conductive features at the bonding surface of the first component;

a second substrate in a second component, the second component having a bonding surface, wherein the bonding surface of the second component has a planarized topography and direct hybrid bonded to the bonding surface of the first component;

a second plurality of electrically conductive features at the bonding surface of the second component and bonded to the first plurality of electrically conductive features;

one or more electrically conductive contact pads disposed within an insulating layer of the second component and below the bonding surface of the second component, the one or more electrically conductive contact pads disposed in an area different from the first plurality of electrically conductive features and the second plurality of electrically conductive features, wherein a footprint of the first substrate is smaller than a footprint of the second substrate; and

one or more secondary openings in the insulating layer of the second component providing electrical access to the one or more electrically conductive contact pads.

21

13. The microelectronic assembly of claim 12, wherein the first substrate comprises a base semiconductor material that has been etched to form at least one island in an active area of the first component, and wherein an insulating layer of the first component between the first substrate and the bonding surface of the first component extends laterally beyond the footprint of the first substrate.

14. The microelectronic assembly of claim 13, further comprising a metal layer overlying at least a portion of the first component on a side opposite the bonding surface of the first component.

15. The microelectronic assembly of claim 13, further comprising one or more primary openings in an insulating layer of the first component, aligned to the one or more secondary openings and to the electrically conductive contact pads, the one or more primary openings extending to the one or more secondary openings, providing electrical access to the one or more electrically conductive contact pads.

16. A microelectronic assembly, comprising:

a first substrate having a bonding surface, the bonding surface of the first substrate having a planarized topography;

a first plurality of electrically conductive features at the bonding surface of the first substrate;

a second substrate having a bonding surface, the bonding surface of the second substrate having a planarized topography and direct hybrid bonded to the bonding surface of the first substrate;

a second plurality of electrically conductive features at the bonding surface of the second substrate bonded to the first plurality of electrically conductive features while misaligned to the first plurality of electrically conductive features by a first extent, wherein the first and second plurality of electrically conductive features together define a footprint perimeter;

one or more electrically conductive contact pads disposed within an insulating layer of the second substrate and below the bonding surface of the second substrate;

one or more secondary openings in the insulating layer of the second substrate aligned to the one or more electrically conductive contact pads, the one or more secondary openings extending from the bonding surface of the second substrate to the one or more electrically conductive contact pads, wherein the secondary openings are laterally within the footprint perimeter; and

one or more primary openings in an insulating layer of the first substrate, misaligned to the one or more secondary openings by the first extent, the one or more primary openings extending to the one or more secondary openings and providing electrical access to the one or more electrically conductive contact pads.

17. The microelectronic assembly of claim 16, further comprising one or more tertiary openings in a base layer of the first substrate, aligned to the one or more primary

22

openings in the insulating layer of the first substrate, the one or more tertiary openings extending from an outside surface of the first substrate to the one or more primary openings, providing electrical access to the one or more electrically conductive contact pads from beyond the outside surface of the first substrate.

18. The microelectronic assembly of claim 17, further comprising one or more electrically conductive structures disposed within one or more of the one or more secondary openings, the one or more primary openings, and the one or more tertiary openings, and electrically coupled to the one or more electrically conductive contact pads.

19. The microelectronic assembly of claim 16, wherein a footprint of the first substrate is smaller than a footprint of the second substrate.

20. A microelectronic assembly, comprising:

a first substrate having a bonding surface, the bonding surface of the first substrate having a planarized topography;

a first plurality of electrically conductive features at the bonding surface of the first substrate;

a second substrate having a bonding surface, the bonding surface of the second substrate having a planarized topography and direct hybrid bonded to the bonding surface of the first substrate;

a second plurality of electrically conductive features at the bonding surface of the second substrate and bonded to the first plurality of electrically conductive features;

one or more electrically conductive contact pads disposed within an insulating layer of the second substrate and below the bonding surface of the second substrate, the one or more electrically conductive contact pads disposed in an area different from the first plurality of electrically conductive features and the second plurality of electrically conductive features;

one or more secondary openings in the insulating layer of the second substrate aligned to the one or more electrically conductive contact pads, the one or more secondary openings comprising cavities extending from the bonding surface of the second substrate to the one or more electrically conductive contact pads, providing electrical access to the one or more electrically conductive contact pads;

at least one metal layer disposed on an outside surface of the first substrate, the metal layer configured for at least one of electromagnetic interference (EMI) protection and heat dissipation; and

a base semiconductor island disposed on the outside surface of the first substrate,

wherein the secondary openings are laterally within a perimeter of the first substrate and the first and second plurality of electrically conductive features are laterally outside of the secondary openings.

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